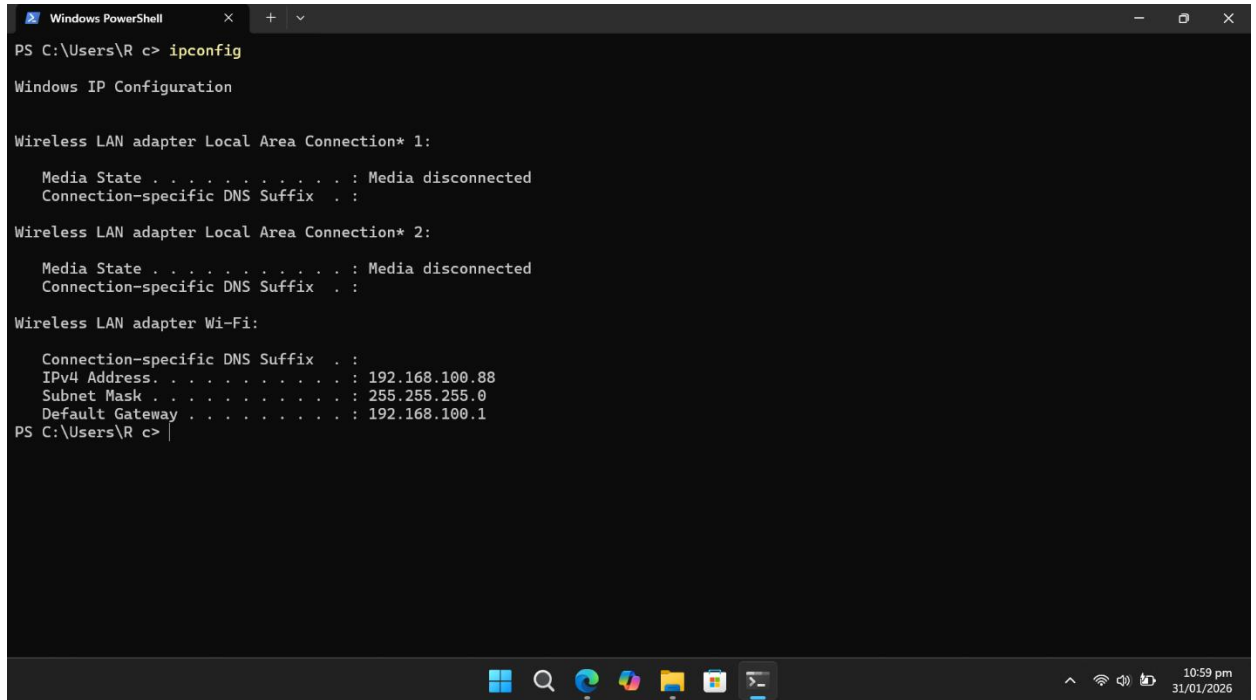


Assignment-01: Windows Network Commands Guide

IP & Network Configuration

- **Ipconfig**



```
Windows PowerShell
PS C:\Users\R c> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

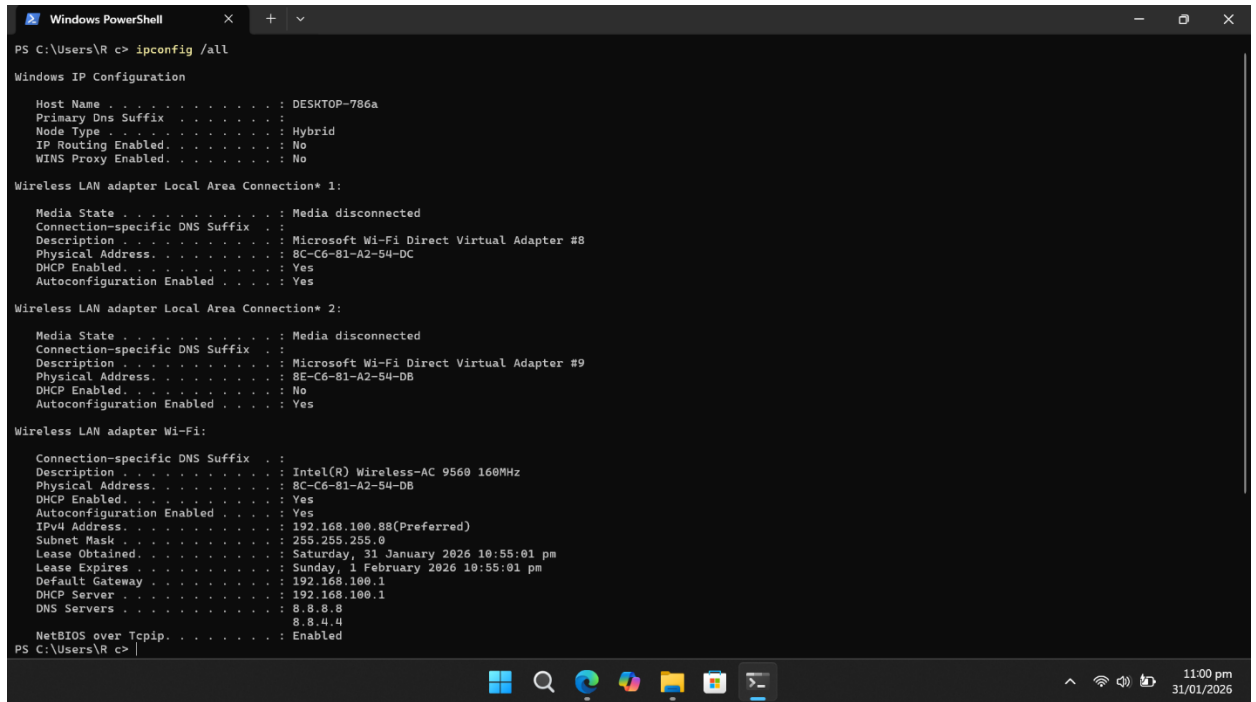
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 192.168.100.88
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1
PS C:\Users\R c> |
```

- **Use:** Displays the basic TCP/IP configuration (IP address, subnet mask, gateway) for active adapters.
- **Working:** Queries the Windows IP stack to retrieve currently assigned parameters for network interfaces.

- **ipconfig /all**



```
Windows PowerShell
PS C:\Users\R c> ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-786a
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #8
Physical Address. . . . . : 8C-C6-81-A2-54-DC
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #9
Physical Address. . . . . : 8E-C6-81-A2-54-DB
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

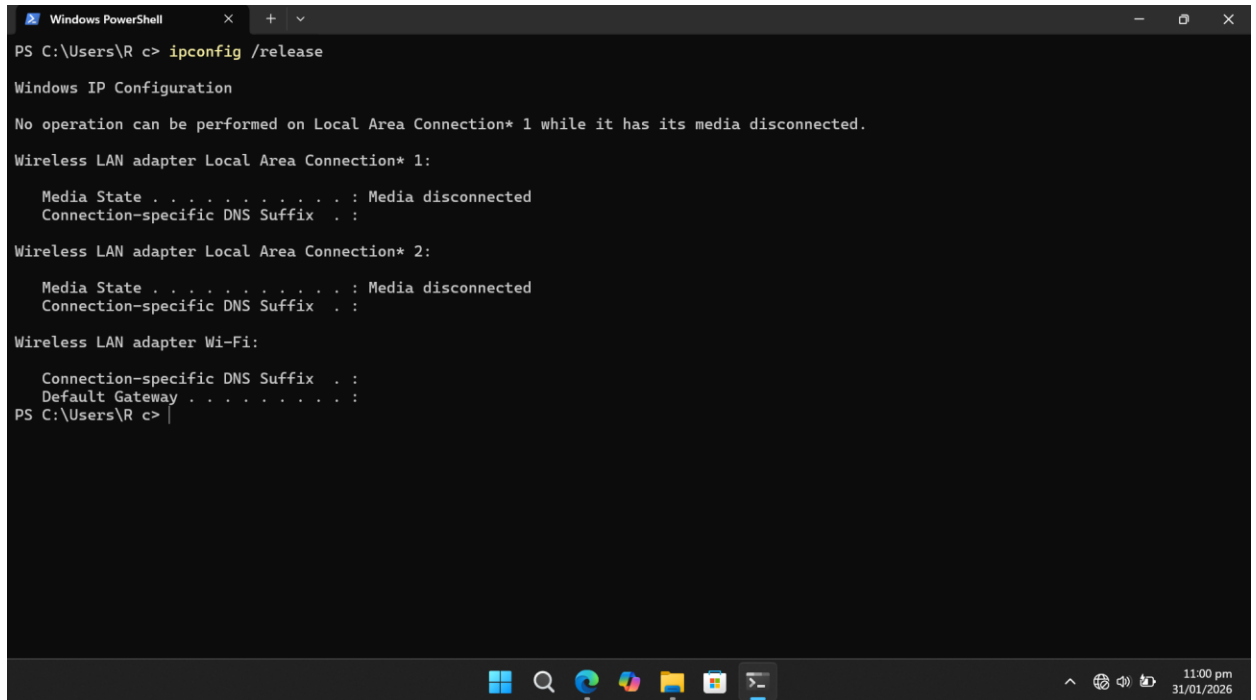
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wireless-AC 9560 160MHz
Physical Address. . . . . : 8C-C6-81-A2-54-DB
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.100.88(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, 31 January 2026 10:55:01 pm
Lease Expires . . . . . : Sunday, 1 February 2026 10:55:01 pm
Default Gateway . . . . . : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DNS Servers . . . . . : 8.8.8.8
                        8.8.4.4
NetBIOS over Tcpip. . . . . : Enabled

PS C:\Users\R c> |
```

- **Use:** Displays detailed configuration for all adapters, including MAC addresses and DHCP status.
- **Working:** Reads extended hardware and driver registry information for every network component, active or inactive.

- **ipconfig /release**



```
Windows PowerShell
PS C:\Users\R > ipconfig /release

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

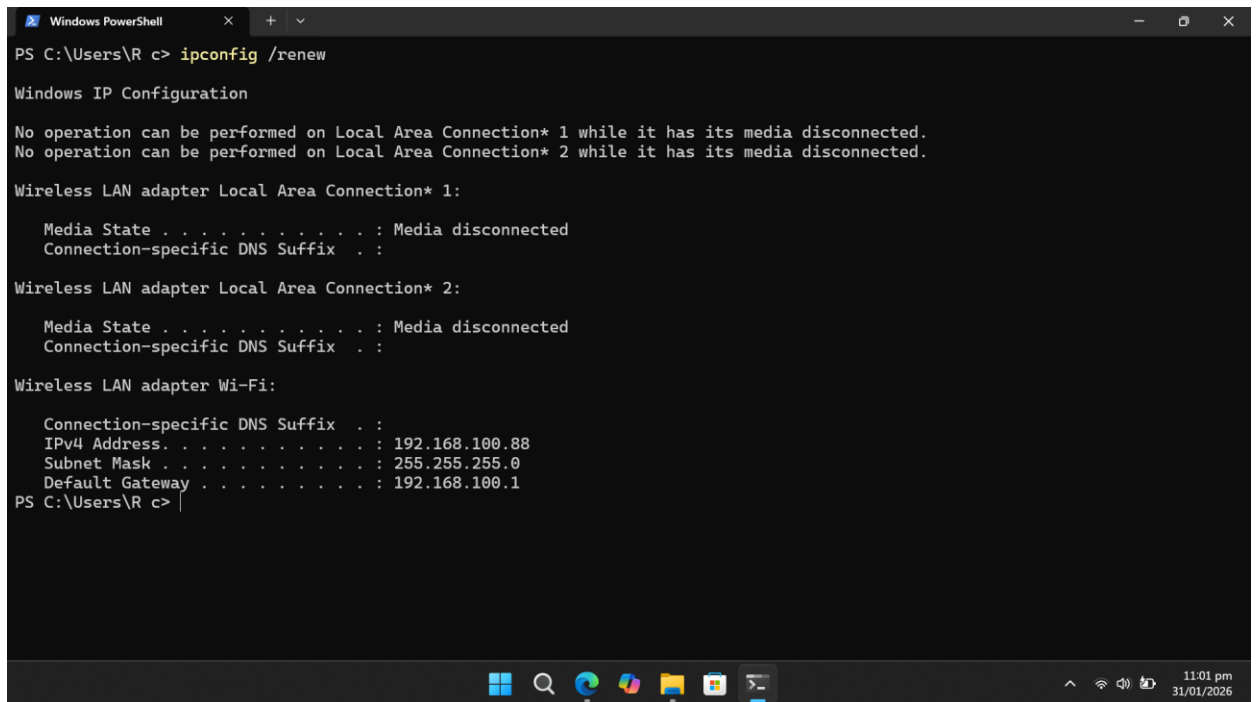
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Default Gateway . . . . . :
PS C:\Users\R > |
```

The screenshot shows a Windows PowerShell window with the command `ipconfig /release` executed. The output displays the status of various network adapters, indicating that they are all disconnected. The taskbar at the bottom shows the time as 11:00 pm on 31/01/2026.

- **Use:** Drops the current IP address assigned by the DHCP server.
- **Working:** Sends a DHCPRELEASE message to the DHCP server, terminating the current lease agreement.

- **ipconfig /renew**



```
Windows PowerShell
PS C:\Users\R c> ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

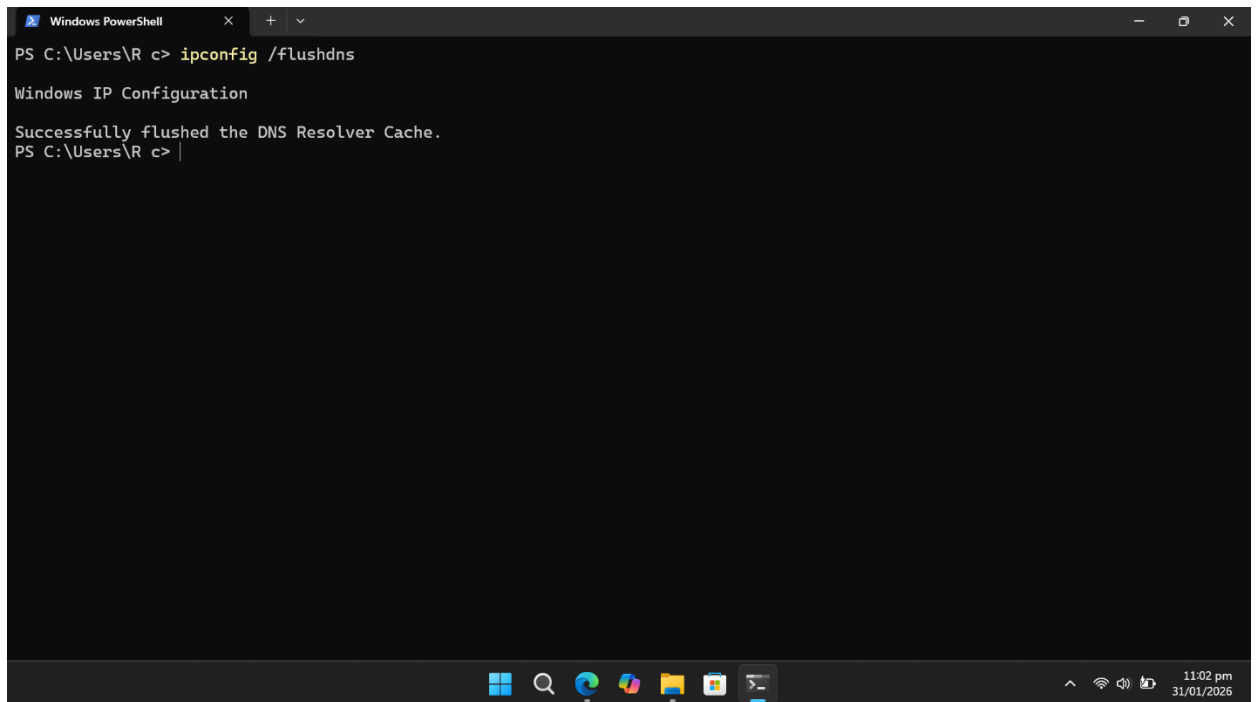
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 192.168.100.88
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1
PS C:\Users\R c>
```

- **Use:** Requests a new IP address from the DHCP server.
- **Working:** Sends a DHCPDISCOVER broadcast to find a server and request a new configuration lease.

- **ipconfig /flushdns**



```
Windows PowerShell
PS C:\Users\R c> ipconfig /flushdns

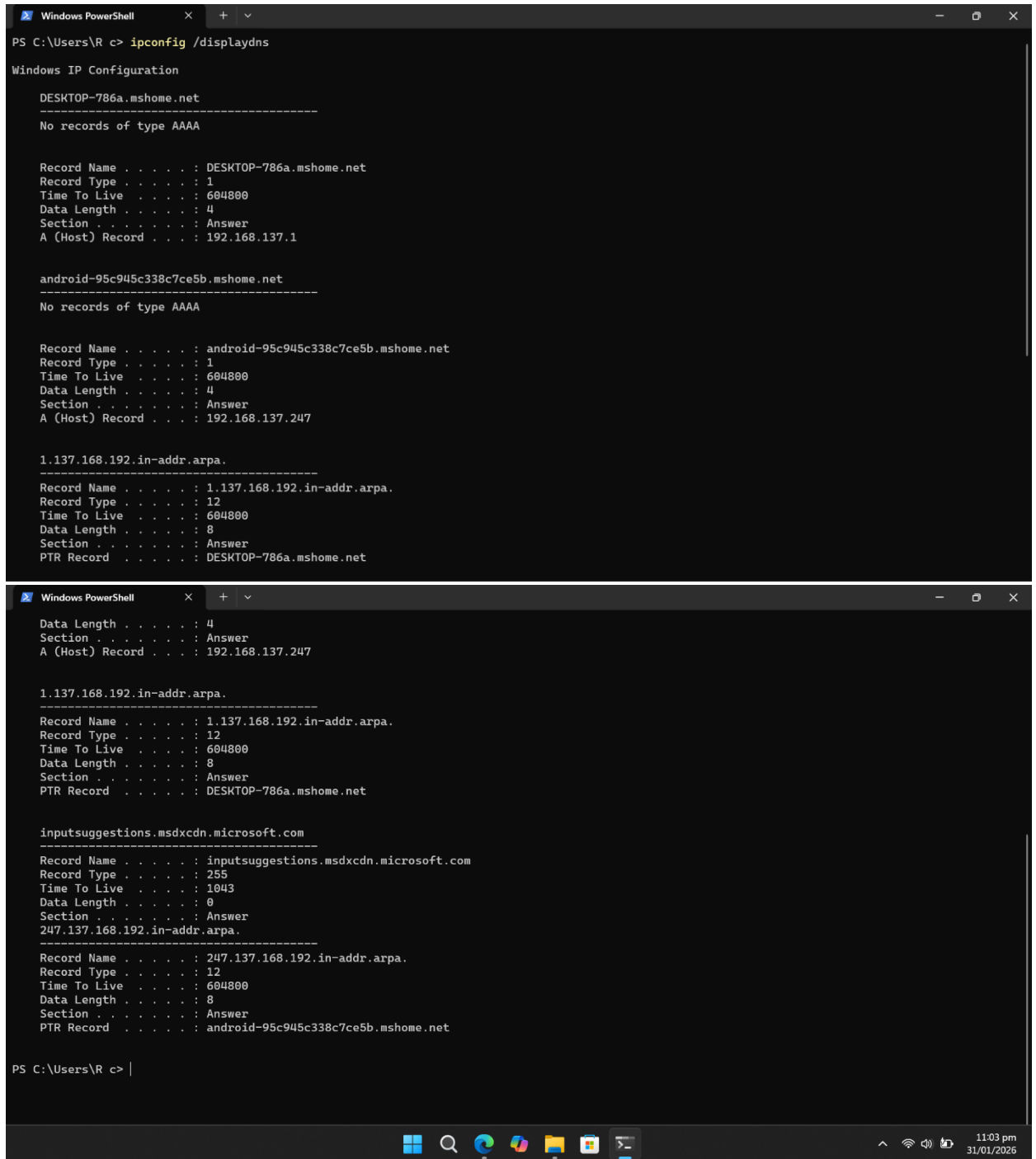
Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
PS C:\Users\R c> |
```

The screenshot shows a Windows PowerShell window with a dark background. The title bar reads 'Windows PowerShell'. The command prompt shows the user 'R' at the 'C:\Users\R' directory. The command 'ipconfig /flushdns' has been entered and executed. The output displays 'Windows IP Configuration' followed by 'Successfully flushed the DNS Resolver Cache.' The prompt is now ready for the next command.

- - **Use:** Deletes the temporary database of visited websites (DNS cache).
 - **Working:** Purges the memory contents of the Windows DNS Client service to force new lookups.

- `ipconfig /displaydns`



```

Windows PowerShell
PS C:\Users\R c> ipconfig /displaydns

Windows IP Configuration

DESKTOP-786a.mshome.net
-----
No records of type AAAA

Record Name . . . . . : DESKTOP-786a.mshome.net
Record Type . . . . . : 1
Time To Live . . . . . : 604800
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 192.168.137.1

-----
android-95c945c338c7ce5b.mshome.net
-----
No records of type AAAA

Record Name . . . . . : android-95c945c338c7ce5b.mshome.net
Record Type . . . . . : 1
Time To Live . . . . . : 604800
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 192.168.137.247

1.137.168.192.in-addr.arpa.
-----
Record Name . . . . . : 1.137.168.192.in-addr.arpa.
Record Type . . . . . : 12
Time To Live . . . . . : 604800
Data Length . . . . . : 8
Section . . . . . : Answer
PTR Record . . . . . : DESKTOP-786a.mshome.net

Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 192.168.137.247

1.137.168.192.in-addr.arpa.
-----
Record Name . . . . . : 1.137.168.192.in-addr.arpa.
Record Type . . . . . : 12
Time To Live . . . . . : 604800
Data Length . . . . . : 8
Section . . . . . : Answer
PTR Record . . . . . : DESKTOP-786a.mshome.net

inputsuggestions.msdxcdn.microsoft.com
-----
Record Name . . . . . : inputsuggestions.msdxcdn.microsoft.com
Record Type . . . . . : 255
Time To Live . . . . . : 1043
Data Length . . . . . : 0
Section . . . . . : Answer
247.137.168.192.in-addr.arpa.
-----
Record Name . . . . . : 247.137.168.192.in-addr.arpa.
Record Type . . . . . : 12
Time To Live . . . . . : 604800
Data Length . . . . . : 8
Section . . . . . : Answer
PTR Record . . . . . : android-95c945c338c7ce5b.mshome.net

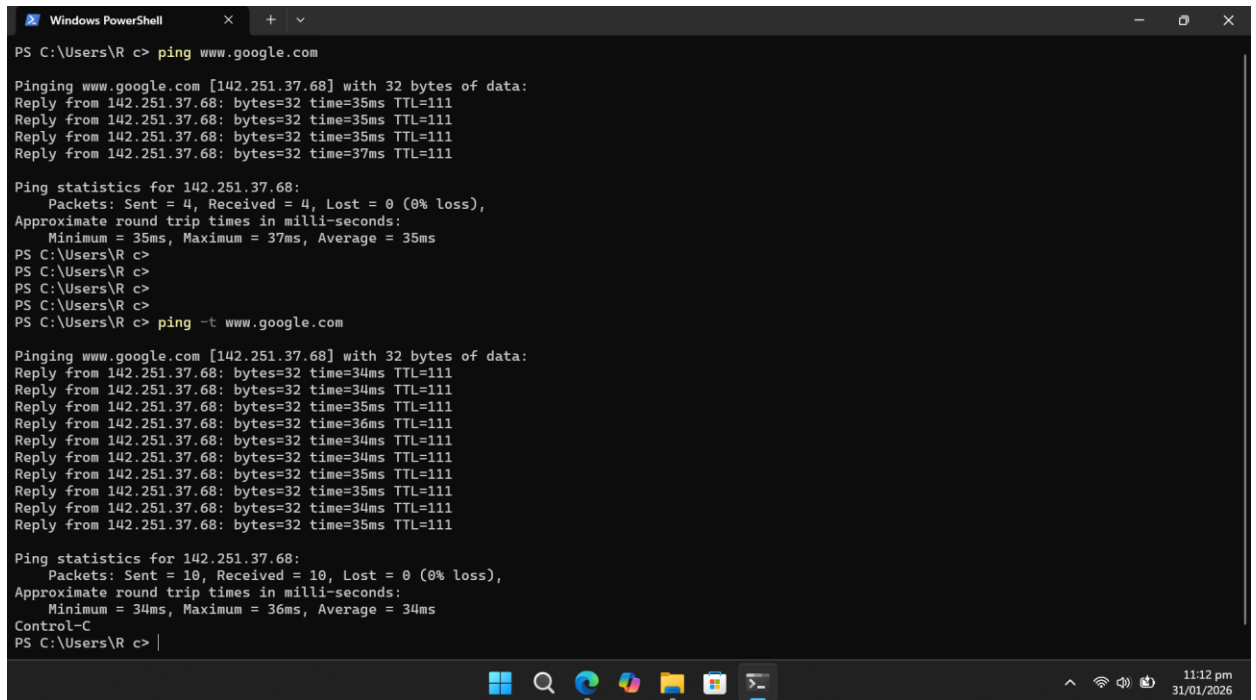
PS C:\Users\R c>

```

- **Use:** Shows the contents of the local DNS resolver cache.
- **Working:** Reads and outputs the list of domain-name-to-IP mappings currently stored in RAM.

Connectivity Testing

- `ping www.google.com` and `ping -t www.google.com`

A screenshot of a Windows PowerShell terminal window. The window title is "Windows PowerShell". The prompt is "PS C:\Users\R c>". The user has entered the command "ping www.google.com". The output shows four successful ping replies from 142.251.37.68 with varying times (35ms, 35ms, 35ms, 37ms) and TTL=111. It then displays ping statistics: 4 packets sent, 4 received, 0% loss, with round trip times ranging from 35ms to 37ms. The user then enters "ping -t www.google.com". The output shows a continuous stream of 10 successful ping replies with times ranging from 34ms to 36ms and TTL=111. It then displays ping statistics: 10 packets sent, 10 received, 0% loss, with round trip times ranging from 34ms to 36ms. The user presses "Control-C" to stop the command. The prompt returns to "PS C:\Users\R c>". The taskbar at the bottom shows the Start button, search icon, and several application icons. The system clock in the bottom right corner shows "11:12 pm 31/01/2026".

```
PS C:\Users\R c> ping www.google.com

Pinging www.google.com [142.251.37.68] with 32 bytes of data:
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=37ms TTL=111

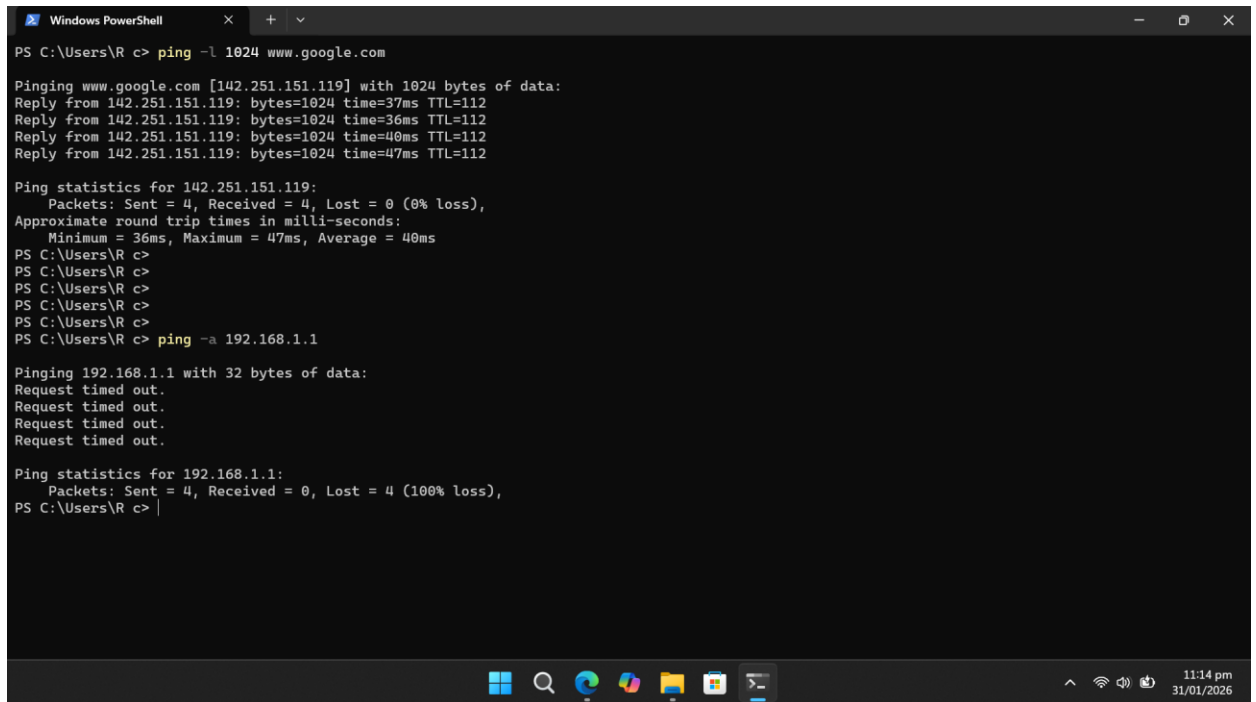
Ping statistics for 142.251.37.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 35ms, Maximum = 37ms, Average = 35ms
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> ping -t www.google.com

Pinging www.google.com [142.251.37.68] with 32 bytes of data:
Reply from 142.251.37.68: bytes=32 time=34ms TTL=111
Reply from 142.251.37.68: bytes=32 time=34ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=36ms TTL=111
Reply from 142.251.37.68: bytes=32 time=34ms TTL=111
Reply from 142.251.37.68: bytes=32 time=34ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111
Reply from 142.251.37.68: bytes=32 time=34ms TTL=111
Reply from 142.251.37.68: bytes=32 time=35ms TTL=111

Ping statistics for 142.251.37.68:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 34ms, Maximum = 36ms, Average = 34ms
Control-C
PS C:\Users\R c> |
```

- **Use:** Tests basic connectivity and latency to a specific host name.
- **Working:** Resolves the name to an IP and sends ICMP Echo Requests, waiting for Echo Replies.
- **Use:** Sends ping requests continuously until manually stopped (Ctrl+C).
- **Working:** loops the standard ping process indefinitely to monitor connection stability over time.

- **ping -l 1024 www.google.com** and **ping -a 192.168.1.1**



```
Windows PowerShell
PS C:\Users\R c> ping -l 1024 www.google.com

Pinging www.google.com [142.251.151.119] with 1024 bytes of data:
Reply from 142.251.151.119: bytes=1024 time=37ms TTL=112
Reply from 142.251.151.119: bytes=1024 time=36ms TTL=112
Reply from 142.251.151.119: bytes=1024 time=40ms TTL=112
Reply from 142.251.151.119: bytes=1024 time=47ms TTL=112

Ping statistics for 142.251.151.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 47ms, Average = 40ms
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> ping -a 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\R c>
```

- - **Use:** Sends ping packets with a specific data size (1024 bytes) instead of the default 32 bytes.
 - **Working:** Pads the ICMP packet payload to test how the network handles larger data frames.
- - **Use:** Finds the hostname associated with a specific IP address.
 - **Working:** Performs a reverse DNS lookup (PTR record query) on the provided IP address.

- **tracert** www.google.com and **pathping** www.google.com

```

Windows PowerShell
PS C:\Users\R c> tracert www.google.com

Tracing route to www.google.com [142.251.151.119]
over a maximum of 30 hops:
  0  *  *  *  Request timed out.
  1  1 ms  *  1 ms 192.168.100.1
  2  3 ms  3 ms 3 ms 100.84.128.1
  3  *  *  *  Request timed out.
  4  36 ms 22 ms 61 ms 172.17.3.54
  5  22 ms 19 ms 21 ms 172.17.3.53
  6  21 ms 21 ms 23 ms 58.181.114.189
  7  21 ms 21 ms 22 ms 10.253.8.42
  8  22 ms 25 ms 21 ms 10.253.4.36
  9  *  *  *  Request timed out.
 10  37 ms 38 ms 46 ms 72.14.219.254
 11  41 ms 41 ms 43 ms 142.251.151.119

Trace complete.
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> pathping www.google.com

Tracing route to www.google.com [142.251.151.119]
over a maximum of 30 hops:
  0  DESKTOP-786a [192.168.100.88]
  1  192.168.100.1
  2  100.84.128.1
  3  *  *  *
Computing statistics for 50 seconds...
Source to Here This Node/Link
Hop RTT Lost/Sent = Pct Lost/Sent = Pct Address
  0 0ms 0/100 = 0% 0/100 = 0% DESKTOP-786a [192.168.100.88]
  1 2ms 0/100 = 0% 0/100 = 0% 192.168.100.1
  2 4ms 0/100 = 0% 0/100 = 0% 100.84.128.1

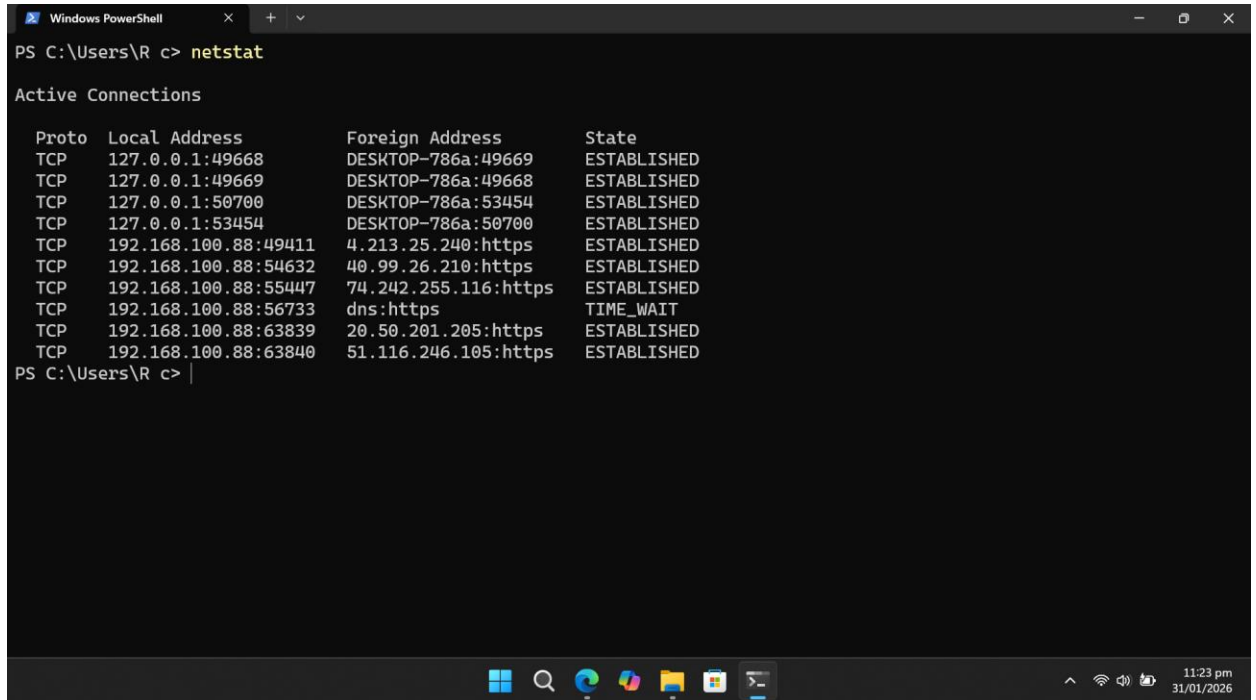
Trace complete.
PS C:\Users\R c>

```

- - **Use:** Shows the path and routers (hops) packets take to reach a destination.
 - **Working:** Sends packets with incrementing TTL (Time-To-Live) values to trigger "Time Exceeded" replies from each router.
 - **Use:** Traces the route and then calculates packet loss statistics for each router.
 - **Working:** Runs a traceroute first, then pings each identified hop multiple times to compute reliability.

Network Statistics & Connections

- **Netstat**



```
PS C:\Users\R c> netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP   127.0.0.1:49668          DESKTOP-786a:49669     ESTABLISHED
TCP   127.0.0.1:49669          DESKTOP-786a:49668     ESTABLISHED
TCP   127.0.0.1:50700          DESKTOP-786a:53454     ESTABLISHED
TCP   127.0.0.1:53454          DESKTOP-786a:50700     ESTABLISHED
TCP   192.168.100.88:49411     4.213.25.240:https     ESTABLISHED
TCP   192.168.100.88:54632     40.99.26.210:https     ESTABLISHED
TCP   192.168.100.88:55447     74.242.255.116:https   ESTABLISHED
TCP   192.168.100.88:56733     dns:https              TIME_WAIT
TCP   192.168.100.88:63839     20.50.201.205:https     ESTABLISHED
TCP   192.168.100.88:63840     51.116.246.105:https   ESTABLISHED

PS C:\Users\R c> |
```

- **Use:** Displays active TCP connections and their current state (e.g., ESTABLISHED).
- **Working:** Queries the system's active socket table to show who is connected to your PC.

- **netstat -an**

```

Windows PowerShell
PS C:\Users\R c> netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP   0.0.0.0:135              0.0.0.0:0               LISTENING
TCP   0.0.0.0:445              0.0.0.0:0               LISTENING
TCP   0.0.0.0:623              0.0.0.0:0               LISTENING
TCP   0.0.0.0:5040             0.0.0.0:0               LISTENING
TCP   0.0.0.0:16992            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49664            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49665            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49666            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49667            0.0.0.0:0               LISTENING
TCP   0.0.0.0:49704            0.0.0.0:0               LISTENING
TCP   127.0.0.1:1001           0.0.0.0:0               LISTENING
TCP   127.0.0.1:49668          127.0.0.1:49669         ESTABLISHED
TCP   127.0.0.1:49669          127.0.0.1:49668         ESTABLISHED
TCP   127.0.0.1:50700          0.0.0.0:0               LISTENING
TCP   127.0.0.1:50700          127.0.0.1:53454         ESTABLISHED
TCP   127.0.0.1:53454          127.0.0.1:50700         ESTABLISHED
TCP   127.0.0.1:56898          0.0.0.0:0               LISTENING
TCP   192.168.100.88:139       0.0.0.0:0               LISTENING
TCP   192.168.100.88:49411     4.213.25.240:443        ESTABLISHED
TCP   192.168.100.88:54632     40.99.26.210:443        ESTABLISHED
TCP   192.168.100.88:55447     74.242.255.116:443      ESTABLISHED
TCP   192.168.100.88:63839     20.50.201.205:443       ESTABLISHED
TCP   192.168.100.88:63840     51.116.246.105:443      TIME_WAIT
TCP   [::]:135                 [::]:0                  LISTENING
TCP   [::]:445                 [::]:0                  LISTENING
TCP   [::]:623                 [::]:0                  LISTENING
TCP   [::]:16992               [::]:0                  LISTENING
TCP   [::]:49664               [::]:0                  LISTENING
TCP   [::]:49665               [::]:0                  LISTENING

TCP   192.168.100.88:55447     74.242.255.116:443      ESTABLISHED
TCP   192.168.100.88:63839     20.50.201.205:443       ESTABLISHED
TCP   192.168.100.88:63840     51.116.246.105:443      TIME_WAIT
TCP   [::]:135                 [::]:0                  LISTENING
TCP   [::]:445                 [::]:0                  LISTENING
TCP   [::]:623                 [::]:0                  LISTENING
TCP   [::]:16992               [::]:0                  LISTENING
TCP   [::]:49664               [::]:0                  LISTENING
TCP   [::]:49665               [::]:0                  LISTENING
TCP   [::]:49666               [::]:0                  LISTENING
TCP   [::]:49667               [::]:0                  LISTENING
TCP   [::]:49704               [::]:0                  LISTENING
TCP   [::]:42050               [::]:0                  LISTENING
TCP   [::]:56898               [::]:0                  LISTENING
UDP   0.0.0.0:500              *:*                      *
UDP   0.0.0.0:4500             *:*                      *
UDP   0.0.0.0:5050             *:*                      *
UDP   0.0.0.0:5353             *:*                      *
UDP   0.0.0.0:5353             *:*                      *
UDP   0.0.0.0:5355             *:*                      *
UDP   0.0.0.0:54458            *:*                      *
UDP   0.0.0.0:56026            2.16.158.43:443         *
UDP   127.0.0.1:44301           *:*                      *
UDP   127.0.0.1:49664          127.0.0.1:49664         *
UDP   192.168.100.88:137       *:*                      *
UDP   192.168.100.88:138       *:*                      *
UDP   [::]:500                 *:*                      *
UDP   [::]:4500                *:*                      *
UDP   [::]:54458               *:*                      *

PS C:\Users\R c>

```

- **Use:** Lists all active connections and listening ports using numbers instead of names.
- **Working:** Displays raw port and IP data from the stack without performing slow DNS name lookups.

- **netstat -r**

```

Windows PowerShell
PS C:\Users\R c> netstat -r
=====
Interface List
16...8c c6 81 a2 54 dc .....Microsoft Wi-Fi Direct Virtual Adapter #8
21...8e c6 81 a2 54 db .....Microsoft Wi-Fi Direct Virtual Adapter #9
24...8c c6 81 a2 54 db .....Intel(R) Wireless-AC 9560 160MHz
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
-----
0.0.0.0                    0.0.0.0          192.168.100.1    192.168.100.88    50
127.0.0.0                  255.0.0.0        On-link          127.0.0.1         331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1         331
127.255.255.255            255.255.255.255  On-link          127.0.0.1         331
192.168.100.0              255.255.255.0    On-link          192.168.100.88    306
192.168.100.88             255.255.255.255  On-link          192.168.100.88    306
192.168.100.255            255.255.255.255  On-link          192.168.100.88    306
224.0.0.0                  240.0.0.0        On-link          127.0.0.1         331
224.0.0.0                  240.0.0.0        On-link          192.168.100.88    306
255.255.255.255            255.255.255.255  On-link          127.0.0.1         331
255.255.255.255            255.255.255.255  On-link          192.168.100.88    306
=====
Persistent Routes:
None

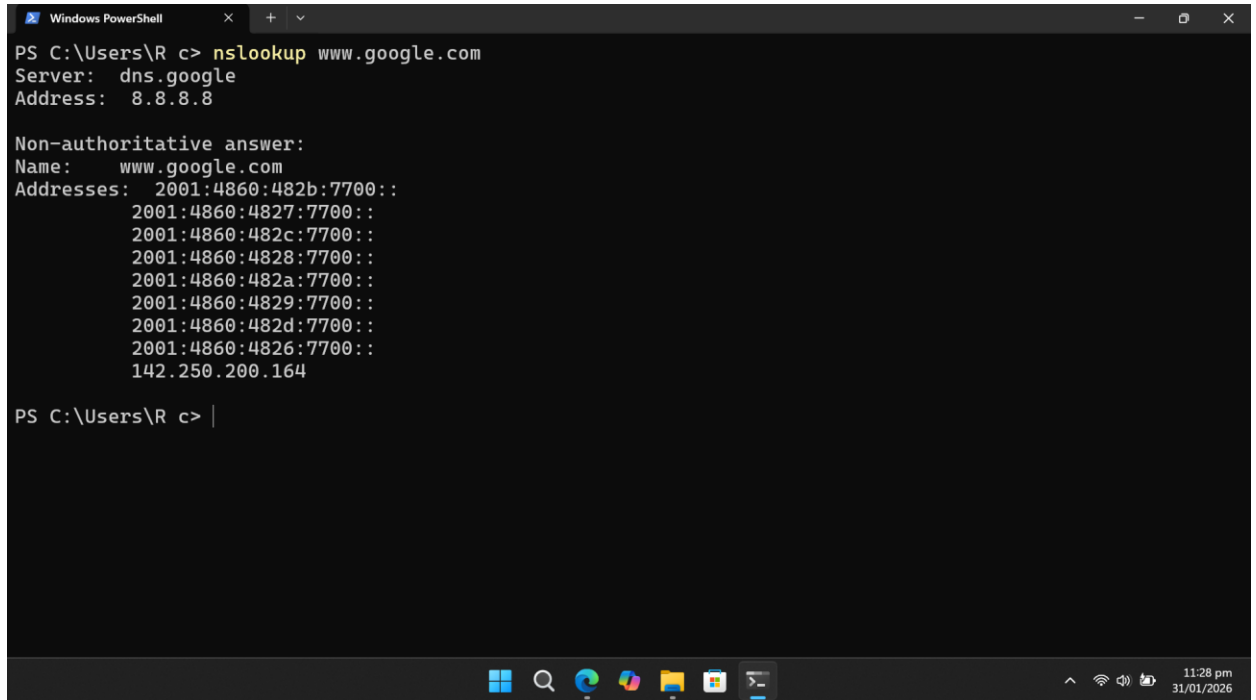
IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
--  -
1    331 ::1/128             On-link
1    331 ff00::/8             On-link
=====
Persistent Routes:
None
PS C:\Users\R c> |

```

- **Use:** Displays the IP routing table.
- **Working:** Shows the kernel's list of network paths used to determine where to send outgoing packets.

DNS & Name Resolution

- nslookup www.google.com



```
Windows PowerShell
PS C:\Users\R c> nslookup www.google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     www.google.com
Addresses: 2001:4860:482b:7700::
           2001:4860:4827:7700::
           2001:4860:482c:7700::
           2001:4860:4828:7700::
           2001:4860:482a:7700::
           2001:4860:4829:7700::
           2001:4860:482d:7700::
           2001:4860:4826:7700::
           142.250.200.164

PS C:\Users\R c> |
```

- **Use:** Queries a DNS server directly to resolve a domain name or IP.
- **Working:** Sends a DNS Query packet to the configured DNS server and displays the raw response.

ARP & Routing

- **arp -a**

```
Windows PowerShell
PS C:\Users\R c> arp -a

Interface: 192.168.100.88 --- 0x18
Internet Address      Physical Address      Type
192.168.100.1         a4-dc-be-7b-67-ca     dynamic
192.168.100.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\R c> |
```

- **Use:** Lists the physical (MAC) addresses resolved for local IP addresses.
- **Working:** Reads the local ARP cache, which stores the hardware addresses of recently communicated devices.

- **route print**

```
Windows PowerShell
PS C:\Users\R c> route print

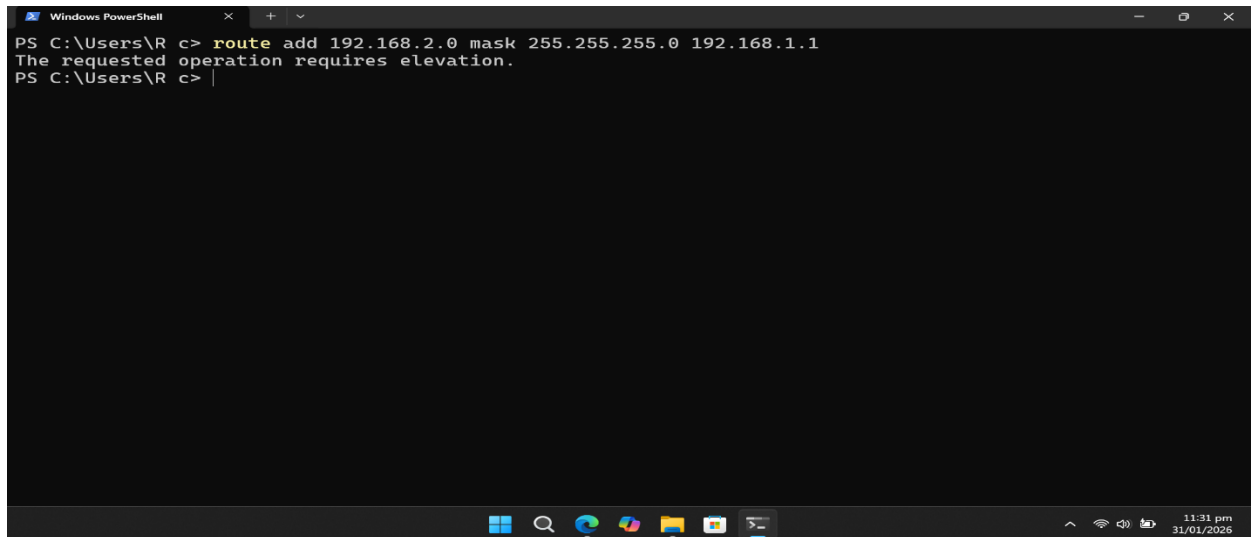
=====
Interface List
16...8c c6 81 a2 54 dc .....Microsoft Wi-Fi Direct Virtual Adapter #8
21...8e c6 81 a2 54 db .....Microsoft Wi-Fi Direct Virtual Adapter #9
24...8c c6 81 a2 54 db .....Intel(R) Wireless-AC 9560 160MHz
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
0.0.0.0                0.0.0.0          192.168.100.1    192.168.100.88   50
127.0.0.0              255.0.0.0        On-link          127.0.0.1        331
127.0.0.1              255.255.255.255  On-link          127.0.0.1        331
127.255.255.255        255.255.255.255  On-link          127.0.0.1        331
192.168.100.0          255.255.255.0    On-link          192.168.100.88   306
192.168.100.88         255.255.255.255  On-link          192.168.100.88   306
192.168.100.255        255.255.255.255  On-link          192.168.100.88   306
224.0.0.0              240.0.0.0        On-link          127.0.0.1        331
224.0.0.0              240.0.0.0        On-link          192.168.100.88   306
255.255.255.255        255.255.255.255  On-link          127.0.0.1        331
255.255.255.255        255.255.255.255  On-link          192.168.100.88   306
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination Gateway
1 331 ::1/128 On-link
1 331 ff00::/8 On-link
=====
Persistent Routes:
None
PS C:\Users\R c> |
```

- **Use:** Displays the entire routing table used by the OS for network traffic decisions.

- **Working:** Outputs the list of active routes, metrics, and interfaces from the TCP/IP stack.
- **route add 192.168.2.0 mask 255.255.255.0 192.168.1.1**

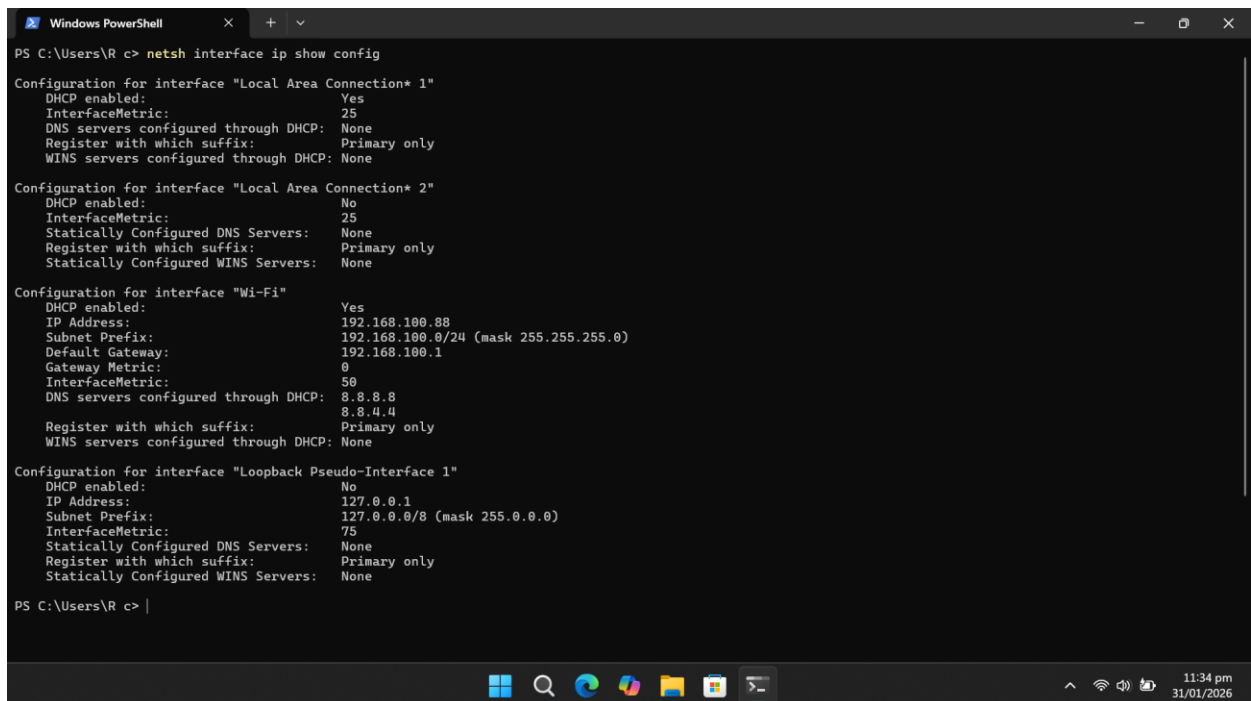


```
Windows PowerShell
PS C:\Users\R> route add 192.168.2.0 mask 255.255.255.0 192.168.1.1
The requested operation requires elevation.
PS C:\Users\R> |
```

- **Use:** Manually adds a static route to the routing table for a specific destination network.
- **Working:** Injects a new rule into the IP routing table directing traffic for the specified subnet to a specific gateway.

Network Interface & Configuration (netsh)

- **netsh interface ip show config**



```
Windows PowerShell
PS C:\Users\R> netsh interface ip show config

Configuration for interface "Local Area Connection* 1"
  DHCP enabled: Yes
  InterfaceMetric: 25
  DNS servers configured through DHCP: None
  Register with which suffix: Primary only
  WINS servers configured through DHCP: None

Configuration for interface "Local Area Connection* 2"
  DHCP enabled: No
  InterfaceMetric: 25
  Statically Configured DNS Servers: None
  Register with which suffix: Primary only
  Statically Configured WINS Servers: None

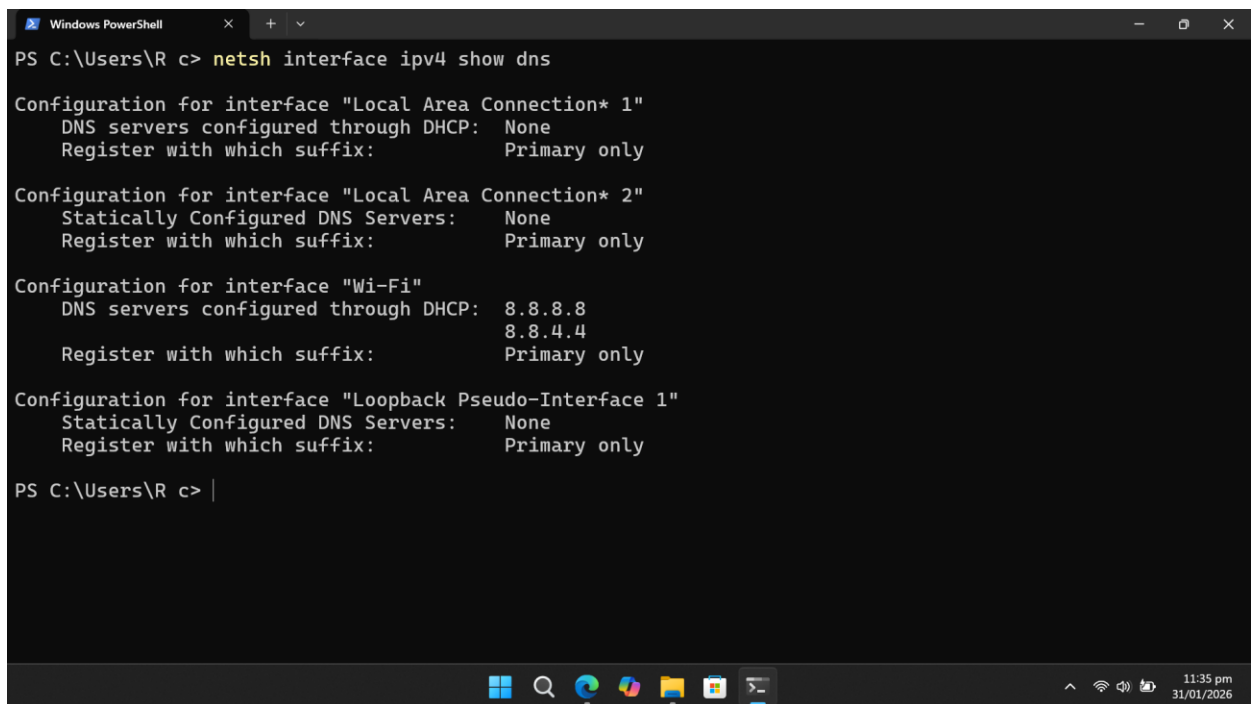
Configuration for interface "Wi-Fi"
  DHCP enabled: Yes
  IP Address: 192.168.100.88
  Subnet Prefix: 192.168.100.0/24 (mask 255.255.255.0)
  Default Gateway: 192.168.100.1
  Gateway Metric: 0
  InterfaceMetric: 50
  DNS servers configured through DHCP: 8.8.8.8, 8.8.4.4
  Register with which suffix: Primary only
  WINS servers configured through DHCP: None

Configuration for interface "Loopback Pseudo-Interface 1"
  DHCP enabled: No
  IP Address: 127.0.0.1
  Subnet Prefix: 127.0.0.0/8 (mask 255.0.0.0)
  InterfaceMetric: 75
  Statically Configured DNS Servers: None
  Register with which suffix: Primary only
  Statically Configured WINS Servers: None

PS C:\Users\R> |
```

- **Use:** Detailed display of IP settings (IP, mask, gateway, metric) for all interfaces.
- **Working:** Accesses the NetShell context to read IP configuration data from the registry.

- **netsh interface ipv4 show dns**



```
PS C:\Users\R c> netsh interface ipv4 show dns

Configuration for interface "Local Area Connection* 1"
  DNS servers configured through DHCP:  None
  Register with which suffix:           Primary only

Configuration for interface "Local Area Connection* 2"
  Statically Configured DNS Servers:    None
  Register with which suffix:           Primary only

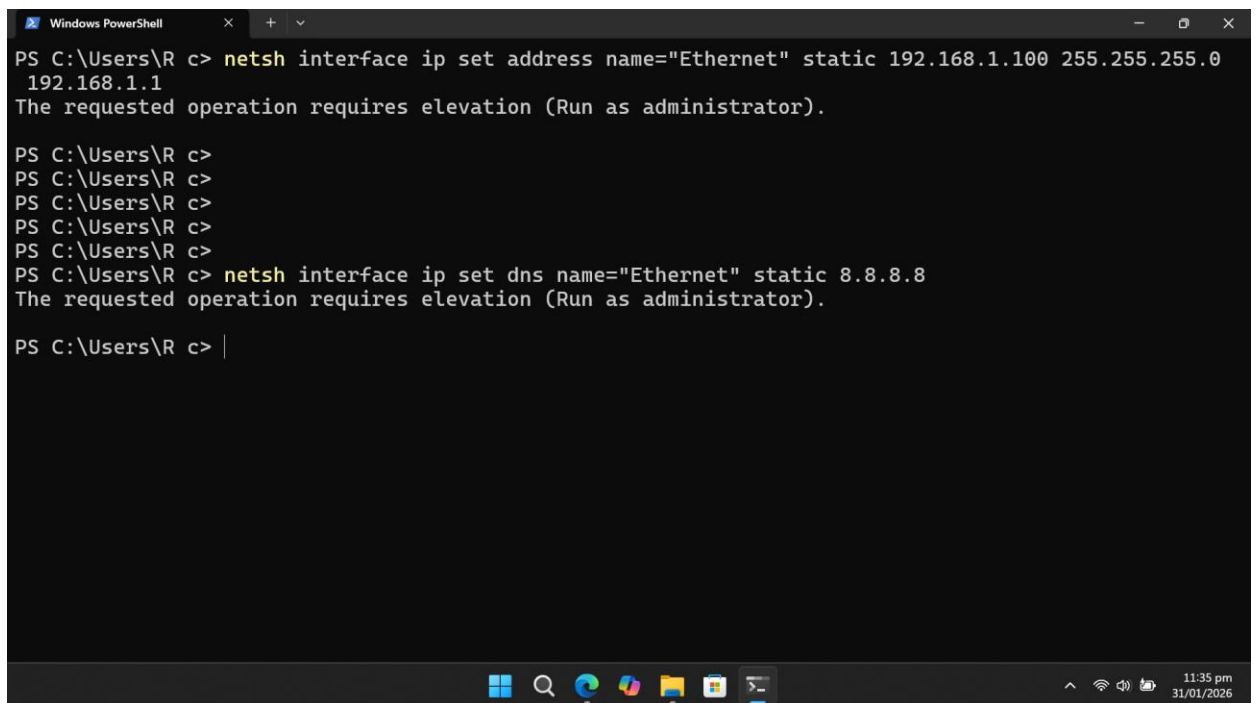
Configuration for interface "Wi-Fi"
  DNS servers configured through DHCP:  8.8.8.8
                                         8.8.4.4
  Register with which suffix:           Primary only

Configuration for interface "Loopback Pseudo-Interface 1"
  Statically Configured DNS Servers:    None
  Register with which suffix:           Primary only

PS C:\Users\R c> |
```

- **Use:** Shows the DNS servers configured for each network adapter.
- **Working:** Retrieves the static or DHCP-assigned DNS server addresses for IPv4 interfaces.

- **netsh interface ip set address name="Ethernet" static 192.168.1.100 ...** and **netsh interface ip set dns name="Ethernet" static 8.8.8.8**



```
Windows PowerShell
PS C:\Users\R c> netsh interface ip set address name="Ethernet" static 192.168.1.100 255.255.255.0 192.168.1.1
The requested operation requires elevation (Run as administrator).

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> netsh interface ip set dns name="Ethernet" static 8.8.8.8
The requested operation requires elevation (Run as administrator).

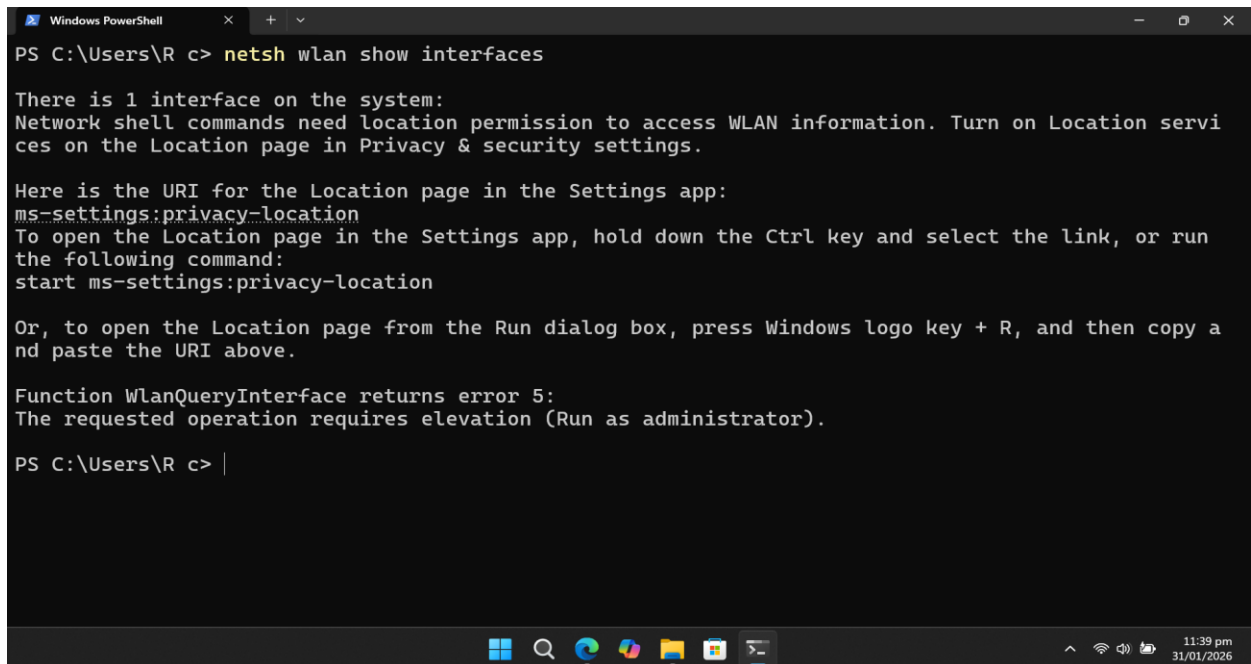
PS C:\Users\R c> |
```

- **Use:** Sets a static IP address, subnet mask, and gateway for a specific adapter.
- **Working:** Disables DHCP on the interface and writes the specified static values to the network configuration.

- **Use:** Manually assigns a specific primary DNS server to an adapter.
- **Working:** Overrides automatic DNS settings and forces the interface to use the specified server IP.

Wireless Networking

- **netsh wlan show interfaces**



```
Windows PowerShell
PS C:\Users\R c> netsh wlan show interfaces

There is 1 interface on the system:
Network shell commands need location permission to access WLAN information. Turn on Location services on the Location page in Privacy & security settings.

Here is the URI for the Location page in the Settings app:
ms-settings:privacy-location
To open the Location page in the Settings app, hold down the Ctrl key and select the link, or run the following command:
start ms-settings:privacy-location

Or, to open the Location page from the Run dialog box, press Windows logo key + R, and then copy and paste the URI above.

Function WlanQueryInterface returns error 5:
The requested operation requires elevation (Run as administrator).

PS C:\Users\R c> |
```

- **Use:** Shows details of the wireless hardware, signal strength, and current connection state.
- **Working:** Queries the WLAN AutoConfig service for the status of the physical Wi-Fi adapter.

- **netsh wlan show profiles**

```

Windows PowerShell
PS C:\Users\R c> netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
-----
<None>

User profiles
-----
All User Profile : WLED-AP
All User Profile : INFINIX NOTE 40
All User Profile : F-11
All User Profile : OPPO A12
All User Profile : Overdosed
All User Profile : Azan
All User Profile : redmi
All User Profile : Sipra Love
All User Profile : Shhhhhhhiiiiiaaaaaooooo
All User Profile : Shhhhhhhiiiiiaaaaaooooo
All User Profile : ESP_44B3AD
All User Profile : W4Ph1_34732
All User Profile : F-11 2
All User Profile : ESP32-Drone
All User Profile : iPhone
All User Profile : ManagementAP
All User Profile : Doctor Abbas
All User Profile : Pixel_1664
All User Profile : Fiber Net I 16
All User Profile : Overdosed 2
All User Profile : C:\Malware\TrojanHorse\Virus.exe
All User Profile : fiber net 1
All User Profile : OVERDOSED
All User Profile : Infenix NOTE 40
All User Profile : ALIENWARE-5G
All User Profile : ALIENWARE

PS C:\Users\R c> |

```

- **Use:** Lists the names (SSIDs) of all Wi-Fi networks saved on the computer.
- **Working:** Reads the stored XML profile files that contain connection details for known networks.

- **netsh wlan connect name="NetworkName"**

```

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\R c> netsh wlan connect name="F-11"
Network shell commands need location permission to access WLAN information. Turn on Location services on the Location page in Privacy & security settings.

Here is the URI for the Location page in the Settings app:
ms-settings:privacy-location
To open the Location page in the Settings app, hold down the Ctrl key and select the link, or run the following command:
start ms-settings:privacy-location

Or, to open the Location page from the Run dialog box, press Windows logo key + R, and then copy and paste the URI above.

Function WlanGetAvailableNetworkList returns error 5:
Access is denied.

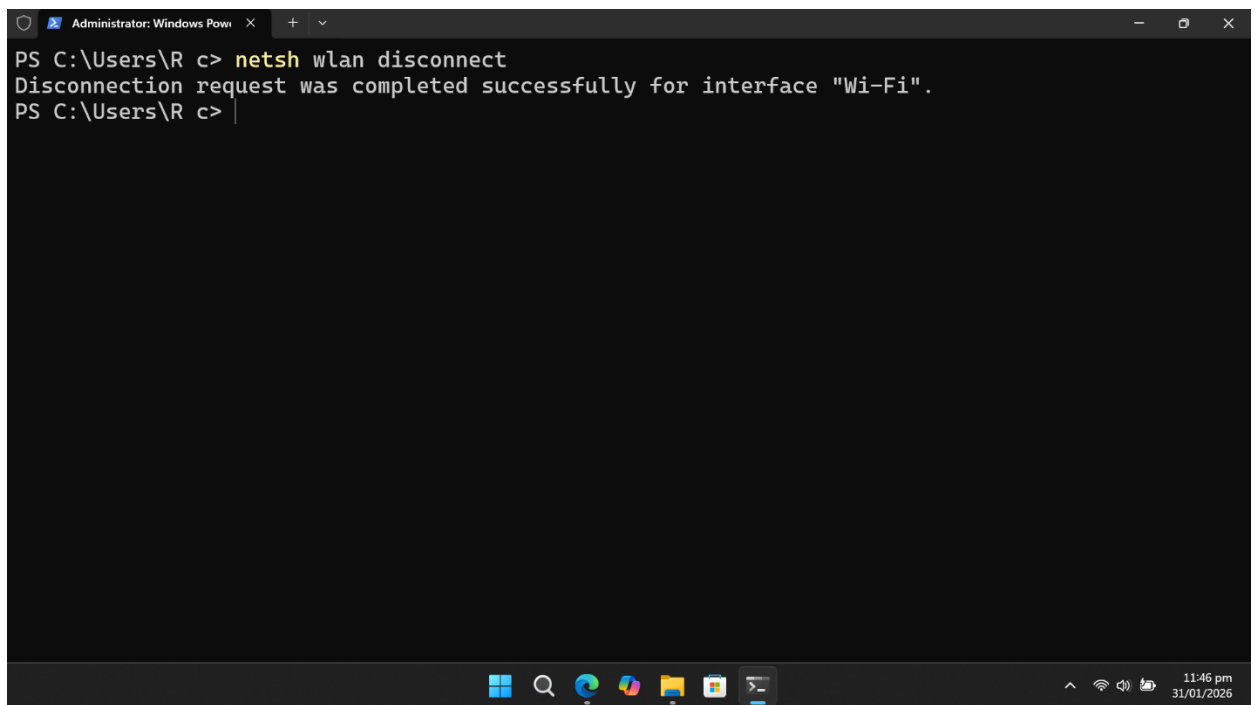
PS C:\Users\R c> |

```

- **Use:** Initiates a connection to a specific saved Wi-Fi network.

- **Working:** Commands the wireless adapter to authenticate and associate with the specified SSID.

- **netsh wlan disconnect**



```
Administrator: Windows PowerShell
PS C:\Users\R c> netsh wlan disconnect
Disconnection request was completed successfully for interface "Wi-Fi".
PS C:\Users\R c>
```

- **Use:** Disconnects the wireless adapter from the current Access Point.
- **Working:** Terminates the radio link layer connection with the current Wi-Fi network.

Firewall Management

- `netsh advfirewall show allprofiles`

```
Administrator: Windows Powe x + v
PS C:\Users\R c> netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

Private Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

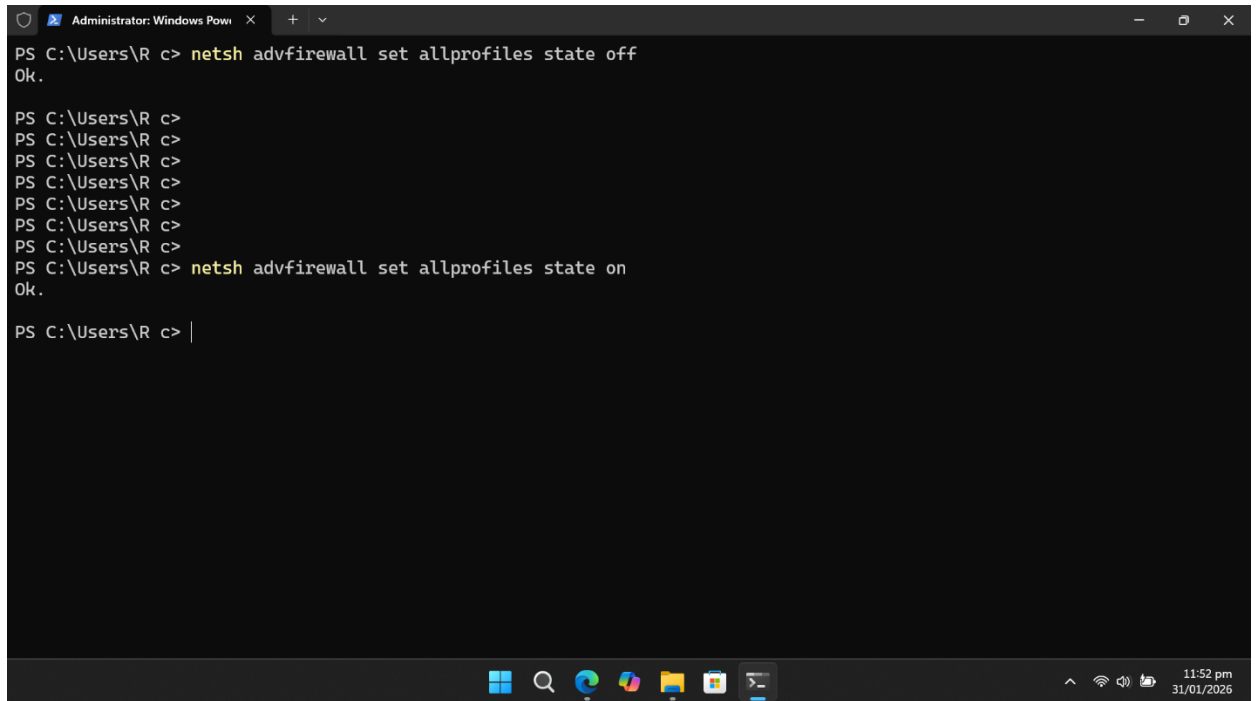
Public Profile Settings:
-----
State                                ON
Firewall Policy                      BlockInbound,AllowOutbound
LocalFirewallRules                   N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

Ok.

PS C:\Users\R c> |
```

- **Use:** Displays the status (ON/OFF) and settings for Domain, Private, and Public firewall profiles.
- **Working:** Reads the current policy configuration of the Windows Defender Firewall.
- **netsh advfirewall set allprofiles state off** and **netsh advfirewall set allprofiles state on**



```
Administrator: Windows PowerShell
PS C:\Users\R c> netsh advfirewall set allprofiles state off
Ok.

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> netsh advfirewall set allprofiles state on
Ok.

PS C:\Users\R c> |
```

- **Use:** Turns off the Windows Firewall for all network profiles (requires Admin).
- **Working:** Modifies the system security policy to bypass packet filtering rules.
- **Use:** Enables the Windows Firewall for all network profiles.
- **Working:** Re-activates packet filtering and rule enforcement for network traffic.

File Sharing & Network Resources

```
Administrator: Windows Powe x + v
PS C:\Users\R c> net use Z: \\servername\sharename
System error 67 has occurred.

The network name cannot be found.

PS C:\Users\R c> net use Z: \\192.168.1.10\sharename
System error 67 has occurred.

The network name cannot be found.

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> net share sharename=C:\folderpath
System error 2 has occurred.

The system cannot find the file specified.

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> net view \\servername
System error 53 has occurred.

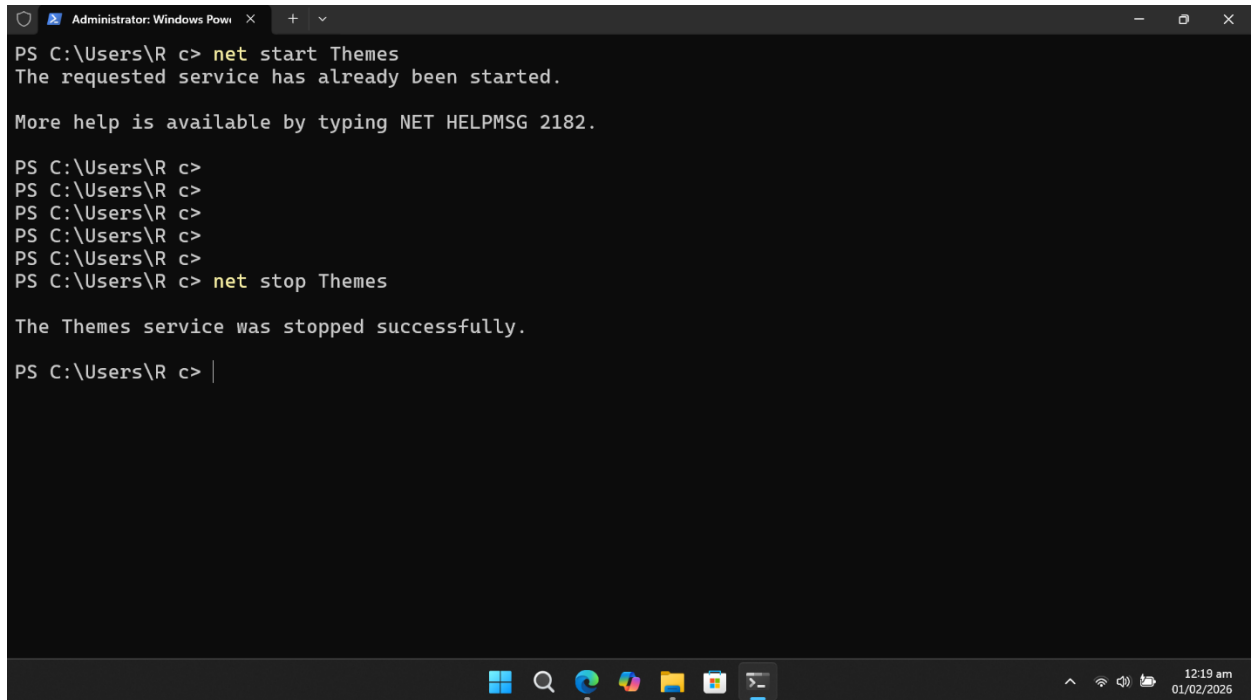
The network path was not found.

PS C:\Users\R c> |
```

- **net use Z: \servername\sharename**
 - **Use:** Maps a network shared folder to a local drive letter (Z:) using the hostname.
 - **Working:** Establishes an SMB session with the server and mounts the share as a virtual drive.
- **net use Z: \192.168.1.10\sharename**
 - **Use:** Maps a network shared folder using the server's IP address directly.
 - **Working:** Connects via SMB using the IP, useful if DNS hostname resolution is failing.
- **net share sharename=C:\folderpath**
 - **Use:** Shares a specific local folder on the network with a given share name.
 - **Working:** Modifies the Server service configuration to allow remote access to the specified directory.
- **net view \servername**
 - **Use:** Lists the shared folders and printers available on a specific computer.
 - **Working:** Queries the remote computer's browse list or SMB share enumeration service.

Network Services & Statistics

- **net start servicename** and **net stop servicename**



```
Administrator: Windows PowerShell
PS C:\Users\R c> net start Themes
The requested service has already been started.

More help is available by typing NET HELPMSG 2182.

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> net stop Themes

The Themes service was stopped successfully.

PS C:\Users\R c> |
```

- **Use:** Starts a specific Windows service (e.g., "net start spooler").
- **Working:** Sends a start command to the Service Control Manager to load and run the service process.
- **Use:** Stops a running Windows service.
- **Working:** Sends a stop signal to the service process, allowing it to close connections and shut down.

- **net statistics workstation**

```
Administrator: Windows Powe x + v
PS C:\Users\R c> net statistics workstation
Workstation Statistics for \\DESKTOP-786A

Statistics since 31/01/2026 10:45:51 pm

Bytes received                                0
Server Message Blocks (SMBs) received         2
Bytes transmitted                             0
Server Message Blocks (SMBs) transmitted      0
Read operations                              0
Write operations                              0
Raw reads denied                             0
Raw writes denied                            0

Network errors                                0
Connections made                             0
Reconnections made                           0
Server disconnects                           0

Sessions started                             0
Hung sessions                                0
Failed sessions                              0
Failed operations                            0
Use count                                     0
Failed use count                             0

The command completed successfully.
PS C:\Users\R c> |
```

- **Use:** Shows usage stats for the client side (bytes sent/received, sessions).
 - **Working:** Reads performance counters from the "Workstation" service (LanmanWorkstation).
- **net statistics server and net config server**

```
Administrator: Windows Powe x + v
PS C:\Users\R c> net statistics server
The syntax of this command is:

NET STATISTICS
[WORKSTATION]

PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c>
PS C:\Users\R c> net config server
Server Name                                \\DESKTOP-786A
Server Comment

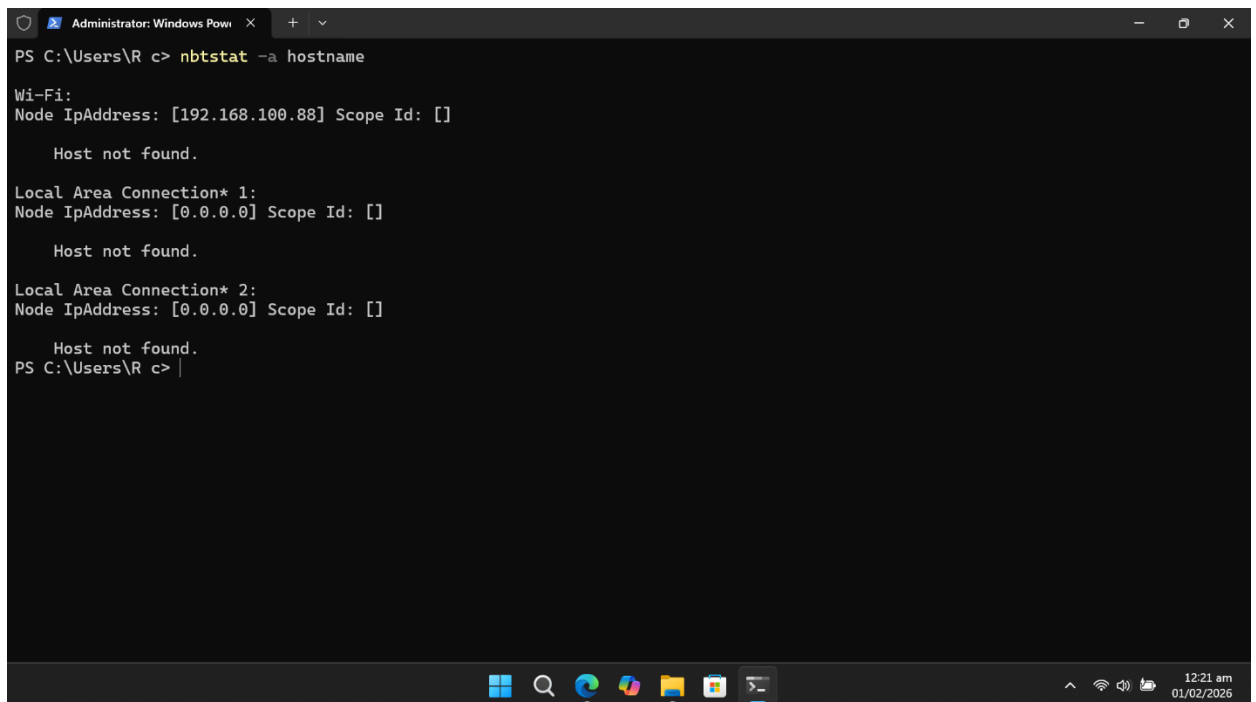
Software version                           Windows 10 Pro
Server is active on
  NetbiosSmb (DESKTOP-786A)
  NetBT_Tcpip_{E6C47289-7612-4129-BAF5-B440C240B81B} (DESKTOP-786A)

Server hidden                               No
Maximum Logged On Users                     20
Maximum open files per session               16384

Idle session time (min)                     15
The command completed successfully.
PS C:\Users\R c> |
```

- **Use:** Shows usage stats for the server side (files opened, permission violations).

- **Working:** Reads performance counters from the "Server" service (LanmanServer).
 - **Use:** Displays and manages configuration for the Server service (like idle session timeout).
 - **Working:** Retrieves the current registry settings governing how the computer shares files.
- **nbtstat -a hostname**



```
Administrator: Windows PowerShell
PS C:\Users\R c> nbtstat -a hostname

Wi-Fi:
Node IpAddress: [192.168.100.88] Scope Id: []

    Host not found.

Local Area Connection* 1:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

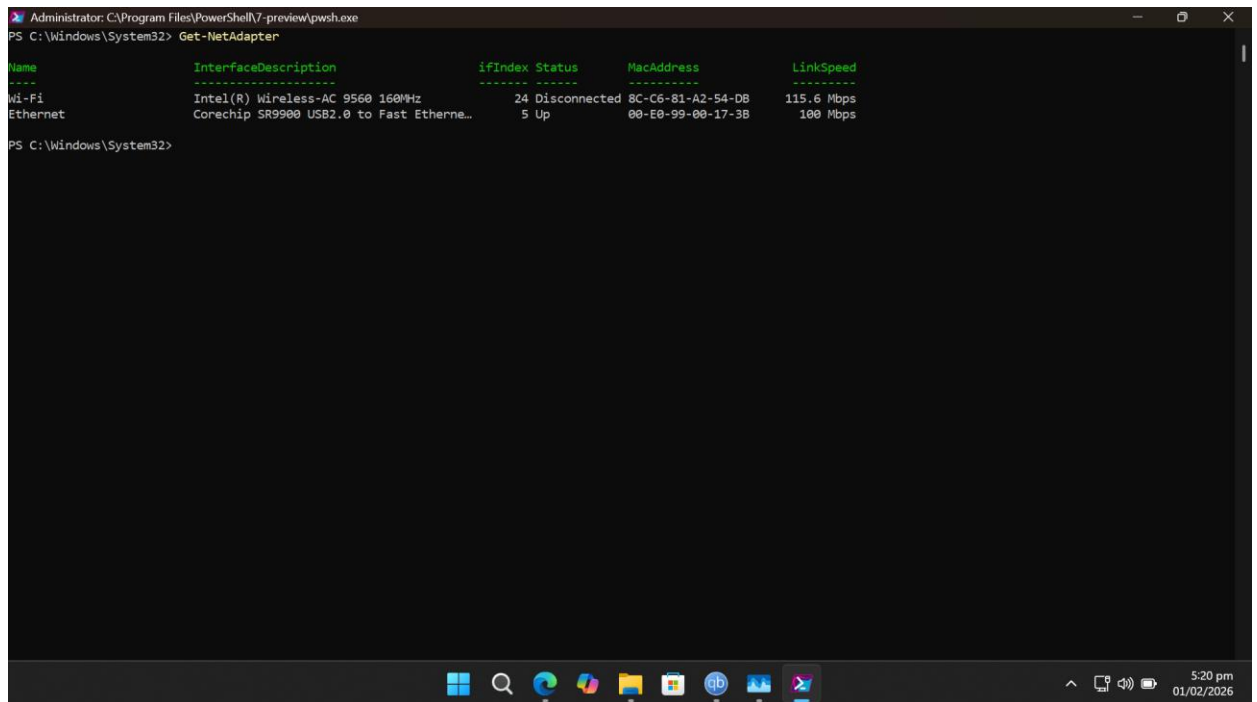
Local Area Connection* 2:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.
PS C:\Users\R c> |
```

- **Use:** Displays the NetBIOS name table of a remote computer.
- **Working:** Sends a NetBIOS Adapter Status Query to the remote machine to get its registered names.

PowerShell Network Commands

- **Get-NetAdapter**



The screenshot shows a Windows PowerShell terminal window titled "Administrator: C:\Program Files\PowerShell\7\powershell.exe". The command "Get-NetAdapter" has been executed, displaying a table of network adapters. The table has five columns: Name, InterfaceDescription, ifIndex, Status, MacAddress, and LinkSpeed. Two adapters are listed: Wi-Fi and Ethernet. The Wi-Fi adapter is Intel(R) Wireless-AC 9560 160MHz, with ifIndex 24, Status Disconnected, MacAddress 8C-C6-81-A2-54-DB, and LinkSpeed 115.6 Mbps. The Ethernet adapter is Corechip SR9900 USB2.0 to Fast Etherne..., with ifIndex 5, Status Up, MacAddress 00-E0-99-00-17-3B, and LinkSpeed 100 Mbps. The prompt "PS C:\Windows\System32>" is visible at the bottom of the terminal.

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
Wi-Fi	Intel(R) Wireless-AC 9560 160MHz	24	Disconnected	8C-C6-81-A2-54-DB	115.6 Mbps
Ethernet	Corechip SR9900 USB2.0 to Fast Etherne...	5	Up	00-E0-99-00-17-3B	100 Mbps

- **Use:** Lists all network adapters (physical and virtual) present in the system.
- **Working:** Retrieves instances of the MSFT_NetAdapter class from WMI/CIM.

- **Get-NetIPAddress**

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> Get-NetIPAddress

IPAddress : ::1
InterfaceIndex : 1
InterfaceAlias : Loopback Pseudo-Interface 1
AddressFamily : IPv6
Type : Unicast
PrefixLength : 128
PrefixOrigin : WellKnown
SuffixOrigin : WellKnown
AddressState : Preferred
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource : False
PolicyStore : ActiveStore

IPAddress : 169.254.179.87
InterfaceIndex : 21
InterfaceAlias : Local Area Connection* 2
AddressFamily : IPv4
Type : Unicast
PrefixLength : 16
PrefixOrigin : WellKnown
SuffixOrigin : Link
AddressState : Tentative
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource : False
PolicyStore : ActiveStore

IPAddress : 169.254.250.79
InterfaceIndex : 16
InterfaceAlias : Local Area Connection* 1
AddressFamily : IPv4
Type : Unicast
PrefixLength : 16
PrefixOrigin : WellKnown
SuffixOrigin : Link
AddressState : Tentative
ValidLifetime : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource : False
PolicyStore : ActiveStore

```

- **Use:** Displays IP address configuration (IPv4 and IPv6) for all interfaces.
- **Working:** Queries the MSFT_NetIPAddress WMI class for current IP settings.

- **Get-NetIPConfiguration and Get-NetRoute**

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> Get-NetIPConfiguration

InterfaceAlias : Ethernet
InterfaceIndex : 5
InterfaceDescription : Corechip SR9900 USB2.0 to Fast Ethernet Adapter
NetProfile.Name : Fiber Net I 16
IPv4Address : 192.168.100.113
IPv4DefaultGateway : 192.168.100.1
DNSServer : 192.168.100.1

InterfaceAlias : Wi-Fi
InterfaceIndex : 24
InterfaceDescription : Intel(R) Wireless-AC 9560 160MHz
NetAdapter.Status : Disconnected

PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Get-NetRoute

ifIndex DestinationPrefix NextHop RouteMetric ifMetric PolicyStore
-----
5 255.255.255.255/32 0.0.0.0 256 35 ActiveStore
21 255.255.255.255/32 0.0.0.0 256 25 ActiveStore
16 255.255.255.255/32 0.0.0.0 256 25 ActiveStore
24 255.255.255.255/32 0.0.0.0 256 50 ActiveStore
1 255.255.255.255/32 0.0.0.0 256 75 ActiveStore
5 224.0.0.0/4 0.0.0.0 256 35 ActiveStore
21 224.0.0.0/4 0.0.0.0 256 25 ActiveStore
16 224.0.0.0/4 0.0.0.0 256 25 ActiveStore
24 224.0.0.0/4 0.0.0.0 256 50 ActiveStore
1 224.0.0.0/4 0.0.0.0 256 75 ActiveStore
5 192.168.100.255/32 0.0.0.0 256 35 ActiveStore
24 192.168.100.255/32 0.0.0.0 256 50 ActiveStore
5 192.168.100.113/32 0.0.0.0 256 35 ActiveStore
24 192.168.100.88/32 0.0.0.0 256 50 ActiveStore
5 192.168.100.0/24 0.0.0.0 256 35 ActiveStore
24 192.168.100.0/24 0.0.0.0 256 50 ActiveStore
1 127.255.255.255/32 0.0.0.0 256 75 ActiveStore
1 127.0.0.1/32 0.0.0.0 256 75 ActiveStore

```

- **Use:** Shows a consolidated view of IP, DNS, and gateway settings (modern ipconfig).

- **Working:** Aggregates data from multiple network CIM objects into a single status view.
- **Use:** Displays the IP routing table (PowerShell equivalent of route print).
- **Working:** Retrieves routing entries from the MSFT_NetRoute WMI class.

Connectivity and DNS

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> Test-Connection www.google.com

Destination: www.google.com

Ping Source          Address          Latency BufferSize Status
-----
1 DESKTOP-786a      142.251.37.68    35      32 Success
2 DESKTOP-786a      142.251.37.68    35      32 Success
3 DESKTOP-786a      142.251.37.68    35      32 Success
4 DESKTOP-786a      142.251.37.68    35      32 Success

PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Resolve-DnsName www.google.com

Name                Type      TTL      Section  IPAddress
-----
www.google.com      AAAA      246      Answer   2a00:1450:4019:818::2004
www.google.com      A         69       Answer   142.251.37.68

PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Get-DnsClientCache

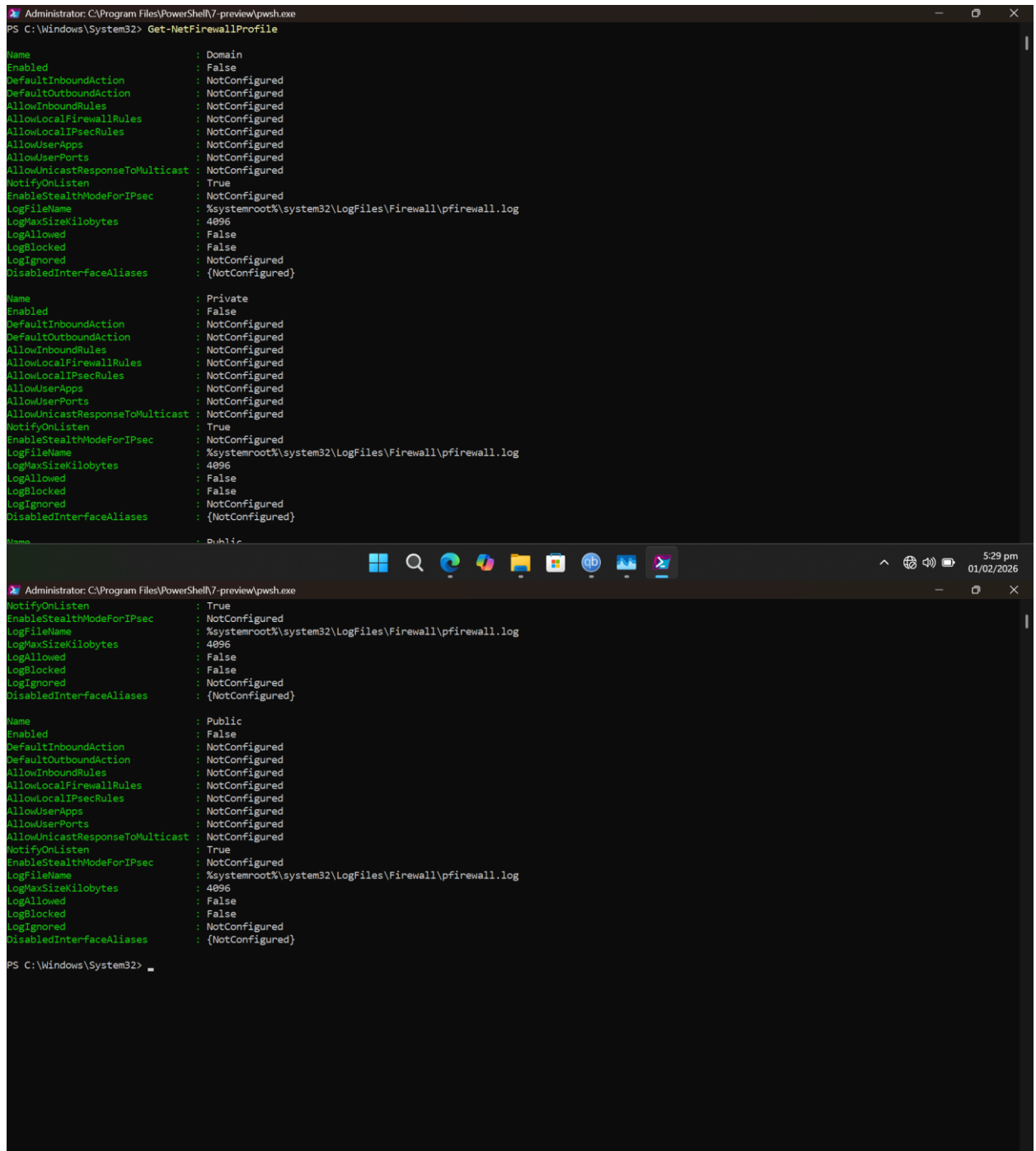
Entry                RecordName      Record  Status  Section  TimeTo  Data  Data
-----                -----
edgedl.me.gvt1.com   edgedl.me.gvt1.com  A       Success Answer    598    4 34.104.35.123
desktop-786a.mshome.net  AAAA      NoRecords
desktop-786a.mshome.net  DESKTOP-786a.mshome.net  A       Success Answer   589101  4 192.168.137.1
android-95c945c338c7ce5b... AAAA      NoRecords
android-95c945c338c7ce5b... android-95c945c338c7ce5b... A       Success Answer   589101  4 192.168.137.247
www.google.com        www.google.com     AAAA      Success Answer    238    16 2a00:1450:4019:818::2004
www.google.com        www.google.com     A         Success Answer     61     4 142.251.37.68
www.google.com        www.google.com     A         Success Answer     61     4 142.251.37.68
tracker.opentrackr.org tracker.opentrackr.org A         Success Answer     39     4 93.158.213.92
p4p.arenabg.com       p4p.arenabg.com    A         Success Answer   2012    4 52.223.39.56
1.137.168.192.in-addr.ar... 1.137.168.192.in-addr.ar... PTR      Success Answer   589101  8 DESKTOP-786a.mshome.net
open.demonii.com      open.demonii.com   A         Success Answer    115     4 185.121.168.96
open.stealth.si       open.stealth.si    A         Success Answer   1260    4 185.243.218.213

```

- **Test-Connection www.google.com**
 - **Use:** Pings a computer to check connectivity (PowerShell equivalent of ping).
 - **Working:** Sends ICMP Echo Requests and returns .NET objects representing the results.
- **Resolve-DnsName www.google.com**
 - **Use:** Performs a DNS lookup to resolve a name to an IP.
 - **Working:** Queries the DNS client to resolve the name, supporting detailed record types.
- **Get-DnsClientCache**
 - **Use:** Displays the contents of the local DNS cache.
 - **Working:** Outputs the currently cached DNS records as objects.

Firewall & Interface Control

- **Get-NetFirewallProfile**



```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> Get-NetFirewallProfile

Name : Domain
Enabled : False
DefaultInboundAction : NotConfigured
DefaultOutboundAction : NotConfigured
AllowInboundRules : NotConfigured
AllowLocalFirewallRules : NotConfigured
AllowLocalIPsecRules : NotConfigured
AllowUserApps : NotConfigured
AllowUserPorts : NotConfigured
AllowUnicastResponseToMulticast : NotConfigured
NotifyOnListen : True
EnableStealthModeForIPsec : NotConfigured
LogFileName : %systemroot%\system32\LogFiles\Firewall\pfirewall.log
LogMaxSizeKilobytes : 4896
LogAllowed : False
LogBlocked : False
LogIgnored : NotConfigured
DisabledInterfaceAliases : {NotConfigured}

Name : Private
Enabled : False
DefaultInboundAction : NotConfigured
DefaultOutboundAction : NotConfigured
AllowInboundRules : NotConfigured
AllowLocalFirewallRules : NotConfigured
AllowLocalIPsecRules : NotConfigured
AllowUserApps : NotConfigured
AllowUserPorts : NotConfigured
AllowUnicastResponseToMulticast : NotConfigured
NotifyOnListen : True
EnableStealthModeForIPsec : NotConfigured
LogFileName : %systemroot%\system32\LogFiles\Firewall\pfirewall.log
LogMaxSizeKilobytes : 4896
LogAllowed : False
LogBlocked : False
LogIgnored : NotConfigured
DisabledInterfaceAliases : {NotConfigured}

Name : Public
Enabled : False
DefaultInboundAction : NotConfigured
DefaultOutboundAction : NotConfigured
AllowInboundRules : NotConfigured
AllowLocalFirewallRules : NotConfigured
AllowLocalIPsecRules : NotConfigured
AllowUserApps : NotConfigured
AllowUserPorts : NotConfigured
AllowUnicastResponseToMulticast : NotConfigured
NotifyOnListen : True
EnableStealthModeForIPsec : NotConfigured
LogFileName : %systemroot%\system32\LogFiles\Firewall\pfirewall.log
LogMaxSizeKilobytes : 4896
LogAllowed : False
LogBlocked : False
LogIgnored : NotConfigured
DisabledInterfaceAliases : {NotConfigured}

PS C:\Windows\System32>
```

- **Use:** Shows the settings for Domain, Private, and Public firewall profiles.
- **Working:** Retrieves the configuration status of the Windows Defender Firewall profiles.

```
Administrator: C:\Program Files\PowerShell\7\powershell.exe
PS C:\Windows\System32> Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Enable-NetAdapter -Name "Ethernet"
Enable-NetAdapter: No MSFT_NetAdapter objects found with property 'Name' equal to 'Ethernet'. Verify the value of the property and retry.
PS C:\Windows\System32> Enable-NetAdapter -Name "Ethernet"
Enable-NetAdapter: No MSFT_NetAdapter objects found with property 'Name' equal to 'Ethernet'. Verify the value of the property and retry.
PS C:\Windows\System32> Enable-NetAdapter -Name "F-11"
Enable-NetAdapter: No MSFT_NetAdapter objects found with property 'Name' equal to 'F-11'. Verify the value of the property and retry.
PS C:\Windows\System32> Enable-NetAdapter -Name "Ethernet"
PS C:\Windows\System32> Disable-NetAdapter -Name "Ethernet"

Confirm
Are you sure you want to perform this action?
Disable-NetAdapter 'Ethernet'
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): Y
PS C:\Windows\System32> _
```

- **Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False**
 - **Use:** Disables the firewall for all three profiles.
 - **Working:** Updates the firewall policy to turn off stateful filtering for the specified profiles.
- **Enable-NetAdapter -Name "Ethernet"**
 - **Use:** Enables a previously disabled network adapter.
 - **Working:** Sends a command to the device driver to power up and activate the interface.
- **Disable-NetAdapter -Name "Ethernet"**
 - **Use:** Disables a network adapter.
 - **Working:** Sends a command to the device driver to stop the interface (administrative down).

Advanced Configuration


```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.100 -PrefixLength 24 -DefaultGateway 192.168.1.1
New-NetIPAddress: Instance MSFT_NetIPAddress already exists
PS C:\Windows\System32>
PS C:\Windows\System32> Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 8.8.8.8
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Get-NetTCPConnection

LocalAddress      LocalPort RemoteAddress      RemotePort State      AppliedSetting OwningProcess
-----
::               49704    ::                0           Listen      Internet      1124
::1             49687    ::                0           Listen      Internet      1368
::             49667    ::                0           Listen      Internet      2828
::             49666    ::                0           Listen      Internet      2020
::             49665    ::                0           Listen      Internet      636
::             49664    ::                0           Listen      Internet      1176
::1            42050    ::                0           Listen      Internet      8712
::            16992    ::                0           Listen      Internet      4128
::1            6586     ::                0           Listen      Internet      11356
::             623     ::                0           Listen      Internet      4128
::             445     ::                0           Listen      Internet      4
::             135     ::                0           Listen      Internet      1492
0.0.0.0         64773    0.0.0.0            0           Bound       Internet      9044
0.0.0.0         63439    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         63433    0.0.0.0            0           Bound       Internet      9044
0.0.0.0         63257    0.0.0.0            0           Bound       Internet      10376
0.0.0.0         60212    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60211    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60210    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60209    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60208    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60206    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60205    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60204    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60202    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60201    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60200    0.0.0.0            0           Bound       Internet      11356
0.0.0.0         60198    0.0.0.0            0           Bound       Internet      11356
192.168.100.88 60198    103.214.160.230    39572      SynSent     Internet      11356
192.168.100.88 60197    79.127.134.20      55517      TimeWait    Internet      0
192.168.100.88 60196    203.76.98.210      6881       SynSent     Internet      11356
192.168.100.88 60195    87.249.133.224     56436      SynSent     Internet      11356
192.168.100.88 58812    185.21.217.31      32774      Established Internet      11356
192.168.100.88 58683    109.134.232.131    48289      Established Internet      11356
192.168.100.88 57260    74.242.255.116     443        Established Internet      9044
127.0.0.1       56986    127.0.0.1          56985      Established Internet      11356
127.0.0.1       56985    127.0.0.1          56986      Established Internet      11356
192.168.100.88 56021    103.180.245.145    3854       TimeWait    Internet      0
192.168.100.88 56006    198.46.168.60      51413      TimeWait    Internet      0
192.168.100.88 55965    146.70.194.54      42486      TimeWait    Internet      0
192.168.100.88 55947    79.127.134.20      55517      TimeWait    Internet      0
192.168.100.88 55921    185.149.91.183     52021      Established Internet      11356
192.168.100.88 55918    185.21.217.33      32769      Established Internet      11356
127.0.0.1       55400    127.0.0.1          64773      Established Internet      1792
127.0.0.1       55400    0.0.0.0            0           Listen      Internet      1792
0.0.0.0         49704    0.0.0.0            0           Listen      Internet      1124
127.0.0.1       49687    0.0.0.0            0           Listen      Internet      1368
127.0.0.1       49669    127.0.0.1          49668      Established Internet      4128
127.0.0.1       49668    127.0.0.1          49669      Established Internet      4128
0.0.0.0         49667    0.0.0.0            0           Listen      Internet      2828
0.0.0.0         49666    0.0.0.0            0           Listen      Internet      2020
0.0.0.0         49665    0.0.0.0            0           Listen      Internet      636
0.0.0.0         49664    0.0.0.0            0           Listen      Internet      1176
192.168.100.88 49442    4.213.25.241       443        Established Internet      11844
192.168.100.88 49441    4.213.25.241       443        Established Internet      11844
0.0.0.0         16992    0.0.0.0            0           Listen      Internet      4128
192.168.100.88 6586     0.0.0.0            0           Listen      Internet      11356
192.168.1.100 6586     0.0.0.0            0           Listen      Internet      11356
127.0.0.1       6586     0.0.0.0            0           Listen      Internet      11356
0.0.0.0         5040     0.0.0.0            0           Listen      Internet      7236
127.0.0.1       1001     0.0.0.0            0           Listen      Internet      4
0.0.0.0         623     0.0.0.0            0           Listen      Internet      4128
192.168.100.88 139      0.0.0.0            0           Listen      Internet      4
192.168.1.100 139      0.0.0.0            0           Listen      Internet      4
0.0.0.0         135      0.0.0.0            0           Listen      Internet      1492
PS C:\Windows\System32>

```

- **New-NetIPAddress -InterfaceAlias "Ethernet" ...**
 - **Use:** Configures a new static IP address, prefix, and gateway on an interface.
 - **Working:** Creates a new MSFT_NetIPAddress instance and binds it to the specified adapter.
- **Set-DnsClientServerAddress -InterfaceAlias "Ethernet" ...**

- **Use:** Sets the DNS server addresses for a specific network interface.
- **Working:** Updates the DNS client configuration for the specified interface index.
- **Get-NetTCPConnection**
 - **Use:** Displays current TCP connections (PowerShell equivalent of netstat).
 - **Working:** Queries the active TCP connection table from the networking stack.

Help & Documentation

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> command /?
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32>
PS C:\Windows\System32> Get-Help Get-NetIPAddress

NAME
    Get-NetIPAddress

SYNTAX
    Get-NetIPAddress [[-IPAddress] <string[]>] [-InterfaceIndex <uint[]>] [-InterfaceAlias <string[]>] [-AddressFamily
    {IPv4 | IPv6}] [-Type {Unicast | Anycast}] [-PrefixLength <byte[]>] [-PrefixOrigin {Other | Manual | WellKnown |
    Dhcp | RouterAdvertisement}] [-SuffixOrigin {Other | Manual | WellKnown | Dhcp | Link | Random}] [-AddressState
    {Invalid | Tentative | Duplicate | Deprecated | Preferred}] [-ValidLifetime <timespan[]>] [-PreferredLifetime
    <timespan[]>] [-SkipAsSource <bool[]>] [-AssociatedIPInterface <CimInstance#MSFT_NetIPInterface[]>] [-PolicyStore
    <string>] [-IncludeAllCompartments] [-CimSession <CimSession[]>] [-ThrottleLimit <int>] [-AsJob]
    [[<CommonParameters>]]

ALIASES
    None

REMARKS
    Get-Help cannot find the Help files for this cmdlet on this computer. It is displaying only partial help.
    -- To download and install Help files for the module that includes this cmdlet, use Update-Help.

PS C:\Windows\System32>
  
```

- **command /?**
 - **Use:** Displays standard help and syntax for a CMD command.
 - **Working:** Invokes the command with the help flag to print its usage instructions.
- **Get-Help Get-NetIPAddress**
 - **Use:** Displays detailed help, syntax, and examples for a PowerShell cmdlet.
 - **Working:** Retrieves help content from the local help files or online documentation.